Indian Institute of Technology Jodhpur
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# Contact No. 1805

## Intellectual Property Disclosure Form

1. **Title of the invention:** A system and method for reconciling multi-source information for patient care in healthcare settings

2. **Innovator(s) who have contributed or conceived an essential element of the invention, either independently or jointly with others during the evolution of the technology concept or reduction to practice:**


(i) Name: Manik Sejwal
Nationality: Indian
Position: Incubatee
Department Address: Medical Technologies. IITJ-AIIMSJ
Phone: +91 9958129899
Email: sejwal.1@iitj.ac.in, manik@easemymed.com
Course: Masters-PhD

(ii) Name: Sumit Kalra
Nationality: Indian
Position: Assistant Professor
Department Address: Department of Computer Science and Engineering, IITJ
Phon: +91 7976777787
Email: sumitk@iitj.ac.in
Course: NA

(iii) Name: Dr. Surender Deora
Nationality: Indian
Position: Additional Professor
Department Address: Department of Cardiology, AIIMSJ
Phone: +91 8003996715
Email: deoras@aiimsjodhpur.edu.in
Course: NA


**Inventors are requested to provide their Full Name (**without initials**), Position (**e.g Professor / Associate Professor / Assistant Professor, Post Doc / PhD / M. Tech / B. Tech Student**), Phone Number (**Personal**), Email ID (**Official along with Personal**)*

**Students are advised to provide their **home address** as well.*

3. Non-Confidential description of the invention in layman's Language:

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### *A. Abstract in 100 words*

A core problem in managing illnesses is that a patient's self-reported information is often incomplete or not sufficient. For example, a patient with diabetes might report perfect adherence, while a caregiver observes them secretly eating sweets. This creates a critical information gap for the doctor.

Our invention is a **system of collaborative AI agents that closes this gap by creating a secure care circle with 3 or more individual users.**

Personal AI agents are assigned to the patient and caregiver(s). These agents:

1. **Gather Information Independently:** Each agent converses naturally with its user, collecting their unique perspective.
2. **Reconcile Conflicting Reports:** The system's central orchestrator compares the information and identifies discrepancies (e.g., "medication_taken: true" vs. "medication_taken: false").
3. **Provide a Unified View:** It generates a more accurate, unified assessment for the doctor, allowing for timely and effective interventions.

This method creates a trusted, mediated communication channel that surfaces the ground truth in complex, real-world health scenarios.

### *B. Use Case*

Social Artificial Intelligence in Healthcare. The system acts as a Care Circle mediator where digital companions for the patient and caregiver communicate to resolve conflicting health reports (e.g., medication adherence, behavioural diagnostics, couple and family therapy) without violating privacy, and provide consolidated dashboards for clinicians.

*Please consult the examples provided in Annexure-1 for filling this section*

### *C. Keywords*

Information and Communications Technology, Social Artificial Intelligence, Healthcare, Multi-agent system, Data reconciliation, Patient adherence, Care circle, Conflict resolution

*Please note that the above keywords will be utilised by the IPR Cell for preparing the Patent Search Report*

**Note:** *Please note that the above Information alone will be circulated to several agencies for technology commercialisation purposes once the patent is filed. Thus, the fields should be self-explanatory to highlight commercialisation potential.*

**4. How does this invention relate to new processes, machines, compositions of matter, etc.? Please cover the following points:**

**(a)** Describe the invention in detail for technical evaluation. Please use additional sheets for sketches, drawing, photographs and other materials that help to illustrate the description.

**Input Devices:** Target and support user devices (e.g., patient and caregiver smartphones) transmitting data streams of health observations.

**Processing Server(s):** Configured with logic modules including a **Multi-Agent Reconciliation Engine** (receives conflicting streams), a **Behavioral State Modeling**

**Engine** (maintains psychological state), and a **Consent Management Engine**. The core logic is distributed among Personal Agents bound to user devices.

**Output Devices:** A clinician's terminal displaying a reconciled data state (e.g., adherence, active conflicts) on a dashboard. The system uses a high-dimensional **State Vector** integrating Clinical, Behavioral, Social, and Conversational states to perform adaptive learning and discrepancy reconciliation.

**(b)** What is Novel in the invention?

The invenion deals wih inerpersonal AI agens. It's core novelty is in the method for mediating and reconciling subjective reports among multiple users and terminals using collaborative AI agents. Unlike prior art, which focused solely on clinician-patient interaction, this system incorporates a third party (caregiver) and uses a "Care Circle" model to identify and resolve discrepancies in self-reported data through automated yet privacy-preserving mediation.

**(c)** What is the "inventive" step in your invention? Is the step non-obvious to a person from related fields?

The inventive step lies in the **privacy-conscious multi-agent reconciliation engine**. The system autonomously identifies factual conflicts (e.g., "caregiver_says_medication_taken: true" vs. "patient_says_medication_taken: false") and generates neutral, mediated clarifying questions to establish "ground truth" without revealing the source of the conflicting report. This involves a distributed network of personal agents performing chain-of-thought processing and privacy discrimination.

**(d)** What are the advantages of the present invention over comparable inventions available in patent literature? Please attach a summary of your patent search*.

**Privacy Preservation:** Enables family counseling and conflict resolution without violating individual privacy by using explicit consent chains.

**Accuracy:** Reconciles multi-source data to fill information gaps common in self-reporting.

**Automation:** Reduces the burden on human doctors by probing conflicting social/behavioural data into a unified view.

**(e)** NOTE: The inventors should go through the Patent Search report carefully and write the difference between his/her invention and each contents of the patent search. For Patent search please contact ipr_rnd@iitj.ac.in

Voss and Raz describe how negotiations involve multiple individuals on the side of both parties, with the implication that both need to be on the same page. They don't describe a technological solution.

Voschina, David, and Jackie Assa. "Personalized care staff dialogue management system for increasing subject adherence of care program." U.S. Patent Application No. 17/564,394. Highlights the sources of information like sensors and details of the chat engine. Limits to patient and health care worker.

Bachrach, Yoram, et al. "Feedback for a conversational agent." U.S. Patent Application No. 15/885,409. - One user converses with the agent. Another type of user identifies the correct response from response options for the query.

Mittal, Ashish, et al. "Multi-agent conversational agent framework." U.S. Patent Application No. 16/552,557. - One user is the seller. Other users are buyers who compete with each other and the agent communicates with the buyers.

Leeds, Richard, and Elon Gasper. "System and method for collaborative conversational AI." U.S. Patent No. 11,431,660. 30 Aug. 2022. - Different chatbots interact with each other and generate a response to a

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human’s query. Similar to bagging of different models.

Aksar, Burak, Yara Rizk, and Tathagata Chakraborti. "Multi-agent chatbot with multi-intent recognition." U.S. Patent Application No. 17/662,484. - Focuses on parsing inputs from a single user.

Nath, Mridul Kumar, and Kingshuk Bose. "Machine-learning recommendation system based on game theory." U.S. Patent Application No. 17/726,297. - Focussed on optimization of video game experience by ensuring that both players get comparable experiences through game theory.

Mohammadzadeh, Niloofar, Reza Safdari, and Azin Rahimi. "Multi-agent system as a new approach to effective chronic heart failure management: key considerations." Healthcare informatics research 19.3 (2013): 162-166. - Doesn’t describe a technical solution and describes a multiagent system like a bagging of models

US-9189749-B2 Knowledge discovery agent system - Chatbot learns as it runs and uncovers information that a user can provide. Data is represented in the form of a knowledge graph. No emphasis on the social aspects of compliance

Balog, Krisztian, and Tom Kenter. "Personal knowledge graphs: A research agenda." Proceedings of the 2019 ACM SIGIR International Conference on Theory of Information Retrieval. 2019.

Personal Knowledge Graphs are tools for making graphs. They include ontologies and other forms of semantic data. They are not, by themselves, meant to be used as chatbots. Don’t create test cases. Are not projected into a spreadsheet.

Sumers, Theodore, et al. "Cognitive Architectures for Language Agents." arXiv preprint arXiv:2309.02427 (2023).

Suggests mentions grounding techniques for cognitive agents. The methods described are for large language models. Dialog graphs and multiplicity of users are not discussed.

Ali, Shaukat, et al. "A systematic review of the application and empirical investigation of search-based test case generation." *IEEE Transactions on Software Engineering* 36.6 (2009): 742-762.
The paper highlights combinatorial testing along with other test case generation methodologies. The proposed methodology is slightly different from ours as we generate scenarios along with the programmed outcome so that it can be verified by the domain expert. We desire identify scenarios that will lead to faults and knowing the fault scenarios will allow us to redesign the behavior. We then validate the same at runtime.

Botsonic -
Trains chatbot around documents.
Trains a large language model on examples mentioned in the document.
A prior dataset is needed.
The proposed methodology makes the user plan and implements the flow.
The proposed method doesn’t use graph-based conversation flows.

Manychat -
Conversation flows are represented as a graph, and each node is a block containing all the information. Blocks are connected to each other and can be rearranged in 2D space.
Graph nodes have messages, options, and connections to the next message.
The UI is not tabular and has excess white space.
Zooming out involves reducing font size
Doesn’t include as many scenarios as proposed by us.
The proposed methodology projects the data into 2 dimensions and aligns equivalent attributes of different nodes.
The proposed methodology focuses on hiding information instead of changing font size.
The proposed methodology takes test case generation into account and presents them in a way that they can be verified.

Omar, Reham, et al. "Chatgpt versus traditional question answering for knowledge graphs: Current status and future directions towards knowledge graph chatbots." *arXiv preprint arXiv:2302.06466* (2023).
Compares Large Language Model (ChatGPT) with a traditional question-answer-based approach to show the superiority of LLM. Our approach doesn't try to outperform LLMs in a single question-answer, but over multiple question-answers using personal conversation graphs.

US20210390431A1 - Knowledge graph weighting during chatbot sessions.
The prior art is not focused on the faults and failures of conversational AI. It is not focused on handcrafting conversation flows. The prior art doesn't tabulate and doesn't include hierarchies. Prior art doesn't focus on creation of anything. Prior art has no concept of collaboration, history and restricted access. Prior art doesn't support the user in creating a structured conversation.

Large language models like ChatGPT and MediSearch now have the knowledge and reasoning ability that surpasses the average human on numerous benchmarks. Despite that, they are still blind to what is going on in the real world and how people behave away from technology. The AI models require wrappers that can collect inputs, which are left out by present technology.

Digital couple and family-based therapy is given through conference calls, whatsapp groups or similar groups controlled by AI (Maia). Privacy and personal companions are excluded.

**(f)** Has the invention been tested experimentally? Are experimental data available?
Yes. Simulation data has been logged. Iterations are also logged in git. Document versions are logged in Google Docs.

**(g)** Technology Readiness Levels (**TRL**) description (mention the applicable stage of TRL given below). Please Mark as Appropriate with adequate justification.

TRL-1 - Research Idea (Potential Application/Basic Principles observed)

TRL-2 - Applied Research Idea (Hypothesis testing and initial proof of concept is demonstrated in a limited number of trials)

TRL-3 - Project Plan (Device Characteristics documents & project proposal completed, Proof-of concept phase)

**TRL-4 - Design and Development** (POC & Safety of device demonstrated by prototype design)

TRL-5 - Standardization (Validating the result of the prototype by testing in simulated environment)

TRL-6 - Preclinical Evaluation (Clinical trials of functional prototype)

TRL-7 - Technology Transfer (Technology transfer of the developed system)

TRL-8 - Clinical Evaluation (Evaluation of the system by clinical trials or demonstration)

TRL-9 - Commercialization (Commercialization & Post Market Surveillance)

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**(h) Need and Demand**

*(Technology gaps addressed in domestic & international markets, pain points of Industry which are being resolved)*

There is a critical information gap in managing illnesses where patient self-reporting is incomplete or inaccurate (e.g., hiding poor adherence). Current solutions like Sensely or Wysa focus on the patient-doctor loop and miss the caregiver's perspective. There is a need for a system that can structure and reconcile these complicated multi-person scenarios. Digital couple and family-based therapy is given through conference calls, whatsapp groups or similar groups controlled by AI (Maia).

**(i) Market Access Information**

(Current Global & domestic Scenario, market size & CAGR)

Global productivity losses due to preventable diseases are estimated at $500 billion.

**(j) Future Developments**

(Scope of future technology development and their application)

core behavioral principles of the Care Circle apply beyond healthcare to domains like Enterprise Sales & Negotiation.

**(k) Application/s of the invention**

(Please refer to Appendix-I)

Social Artificial Intelligence in Healthcare (Patient adherence monitoring, elderly care).
Enterprise Sales & Negotiation (Mediating subjective reports in teams).
Crime investigations

**5. IPR Ownership**

**(a)** Was the intellectual property created with the significant use of funds or facilities of IITJ?

No

**(b)** Please describe any source of funding for the invention (Name of the funding agency and copy of agreement, letter of intent if any, must be enclosed with this form). Bootstrapping, Startup resources (AarogyalinQ Private Limited), Personal devices.

**(c)** What is the source of Salary/Remuneration of inventor/Co-inventor?

Manik - Biodesign Fellow (DBT, AIIMS-IIT)
Dr. Sumit Kalra - IITJ
Dr. Surender Deora - AIIMS

**(d)** Have you presented in any conference, seminar, etc., if yes, please give details?

No. The problem and basic idea were presented during SOTA (December 2023), but the novelty steps were identified, documented and implemented in 2024/2025. They have not been made public.

**(e)** Have you published full/part of this invention, if yes, please give copy of publications?

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No

**(f)** Was the intellectual property created in the course of or pursuant to a sponsored/consultancy research agreement with IITJ? If yes, please enclose a copy of MOU with concerned project.
IIT-AIIMS MOU, Biodesign MOU with DBT, AarogyalinQ-TISC Incubation Agreement

**(g)** Was the intellectual property created as a part of academic research leading towards a degree or otherwise?
Yes

**(h) REVENUE SHARING AMONG INVENTORS:** Please disclose the extent of contribution of each inventor in the invention in percentage terms for revenue sharing.

| NAME OF THE INVENTOR | % SHARE* | SIGNATURE |
|---|---|---|
| 1. **Manik Sejwal** | | |
| 2. **Dr. Sumit Kalra** | | |
| 3. **Dr. Surender Deora** | | |

* If this column is not filled and signed then it will be assumed that all inventor(s) have equal contribution

## 6. Commercial potential

Give brief description of potential commercialization by specifying

**a)** Why should the individual(s)/organization may consider procuring this innovation?
It addresses the unmet need of verifying patient adherence and behavioural diagnosis without invasive measures, providing reliable data for doctors to make effective interventions. This has huge potential in specialisations like mental health as well as medicine.

**b)** These questions are related to the question (i) above:

i. In your opinion what are the steps/processes must be undertaken by the procurer to commercialize the use of this innovation?
AarogyalinQ is already incorporated and incubated. The business plan was presented during incubation selection.
Clinical studies will be required and multiple certifications for data security, ai safety, etc will be required.

ii. How long may it take to reach the commercial stage by the procurer?
Healthcare entry may require 2-3 years but in non-healthcare industries, entry can be as short as 3 months

**c)** Please give specific list of companies and contact details of concerned person who can be contacted for initiating Technology Licensing
AarogyalinQ Private Limited

| | Name of Companies | Name of the contact person | Contact no. |
|---|---|---|---|
| | AarogyalinQ Private Limited | Manik Sejwal | +91 9958129899 |
| | Finplex Solutions | Yashraj Singh | +91 7737114933 |
| | | | |
| | | | |
| | | | |

(**Unsigned & Incomplete IPDF forms shall not be accepted*).

**d)** Do you want to file Patent under PCT Route in other countries?

**Yes** ~~No~~

*PCT & Foreign filing is subject to monetary support from the inventor(s) Project / Personal Account. If the above option is marked "yes", kindly provide the Project Account Details from which the filing cost could be reimbursed.

Project Account Name: AarogyalinQ Private Limited

Project Account No.: 50200103336622

***The institute shall file patent under PCT route only in those cases wherein industry/company has exhibited interest for commercialization.**

**Disclaimer:** *I/We declare that before the submission of this disclosure form or/and during the process of filing this invention as an IPR prospect, I/We will not publish the above information in public domain.*

*I/We also give consent to IIT Jodhpur being the applicant of this IPR prospect, that they may use this disclosure upon their discretion, which will not be limited to publication on e-auction website, Industry meets & different portals for promotional & licensing purposes.*

Sejwal

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11/12/2025

Signature of Inventor with date

Signature of Inventor with date

Signature of Inventor with date

Signature of Inventor with date

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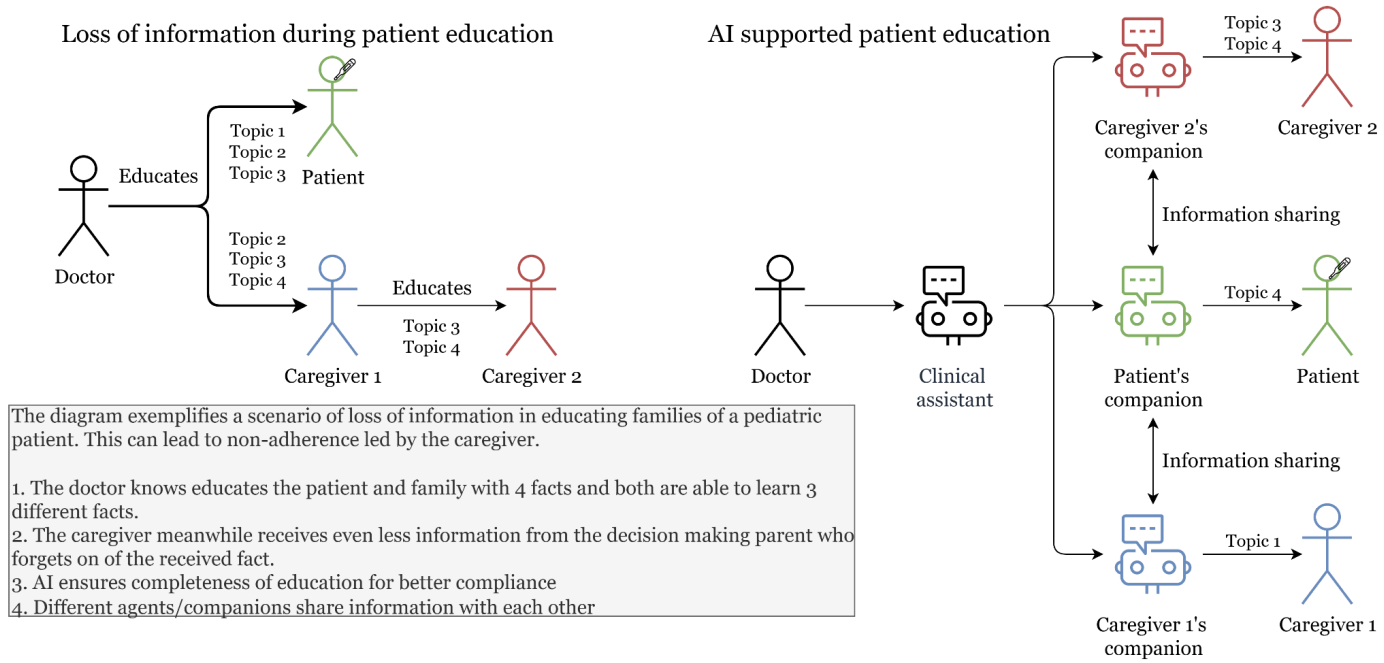



**Fig. 1**: Information gets shared with every member of the patient's team. The digital tools have their copies given to different family members and the tools share information among each other. Like this, it is easier to keep track on any family member who fails to comply. This diagram illustrates the conceptual flow of information. In the preferred embodiment of the invention, all inter-agent communication and reporting to the doctor is mediated through a central server and dashboard, as shown in the main system diagram

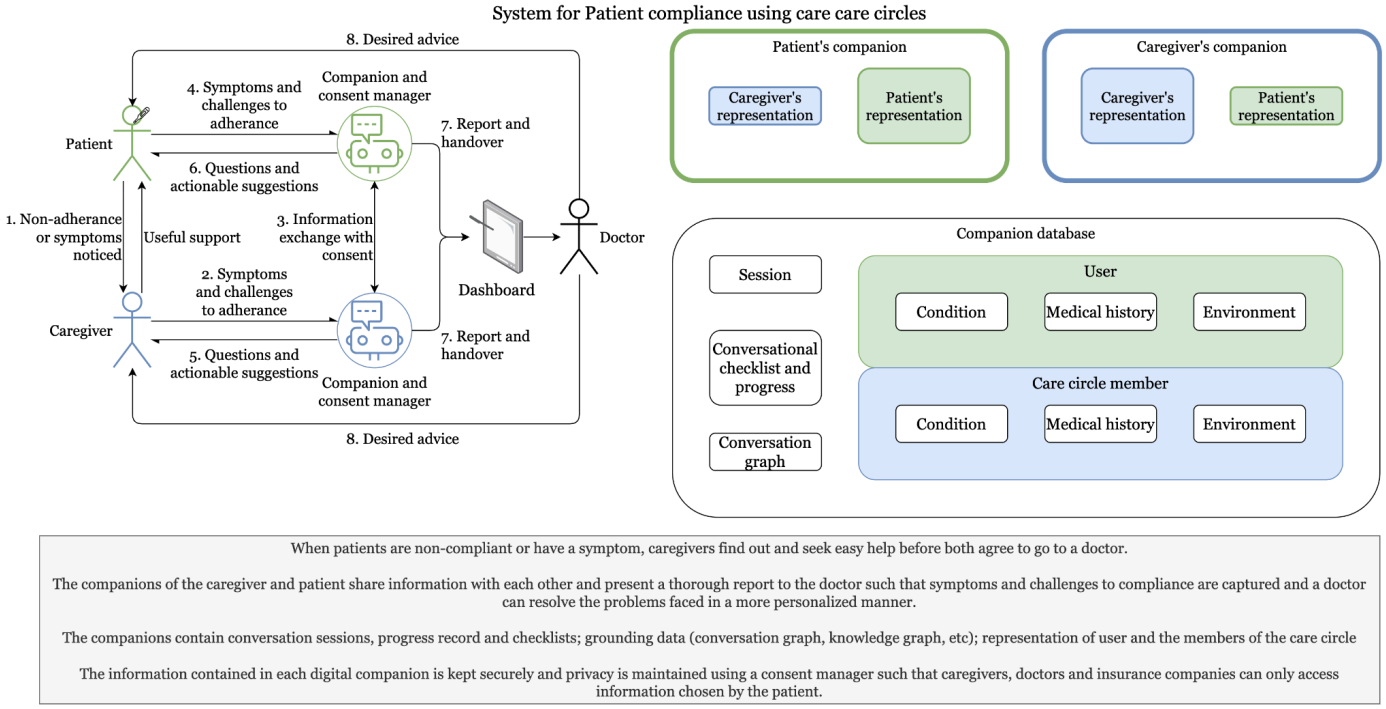



**Fig 2**: This diagram illustrates the overall system architecture and the primary interaction loop between the Patient, Caregiver, and Doctor. The 'Companion and consent manager' represents the system's core logic, the internal workings of which are detailed in subsequent figures

When patients are non-compliant or have a symptom, caregivers find out and seek easy help before both agree to go to a doctor.

The companions of the caregiver and patient share information with each other and present a thorough report to the doctor such that symptoms and challenges to compliance are captured and a doctor can resolve the problems faced in a more personalized manner.

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The companions contain conversation sessions, progress record and checklists; grounding data (conversation graph, knowledge graph, etc); representation of user and the members of the care circle

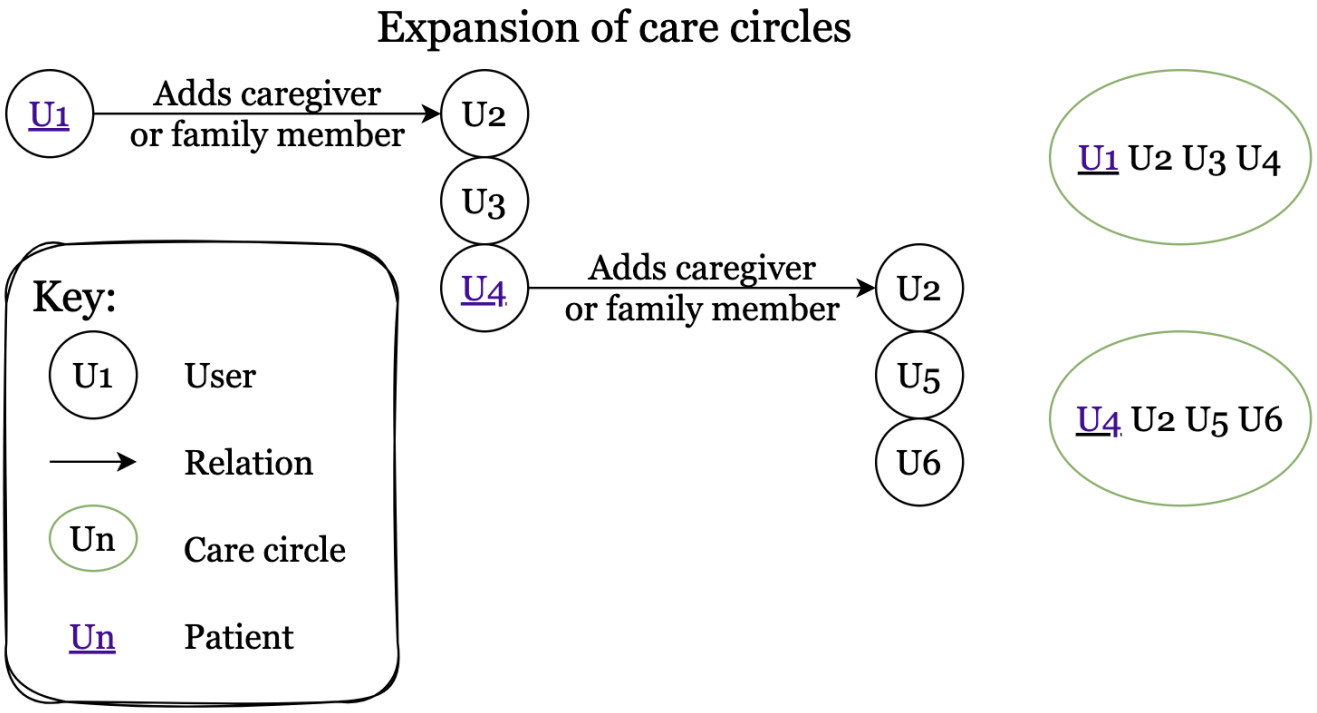



**Fig 3**: Patients add caregivers who they want to share information with

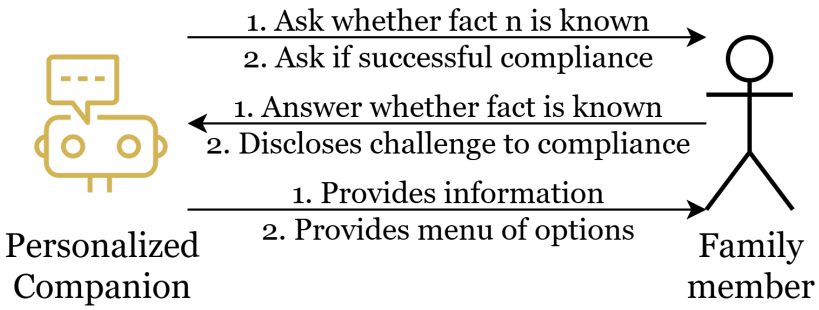



**Fig 4**: The family member discloses information to his or her own copy of the digital tool. Hardware and processing are not included in this diagram

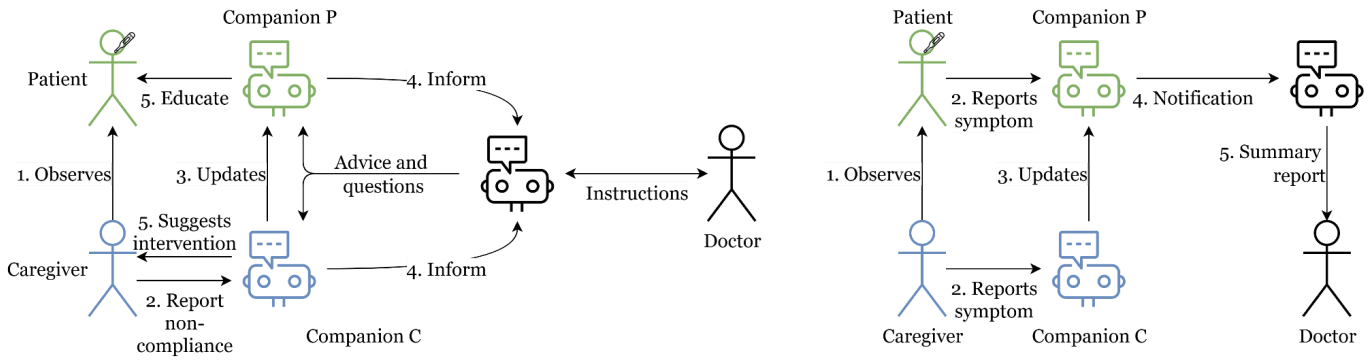



**Fig 5**: The process of sharing information between digital tools and humans has use in patient compliance, as well as complex diagnostic cases. The doctor's digital tool has not been included to highlight the roles of the patient and caregiver, as well as their agents.

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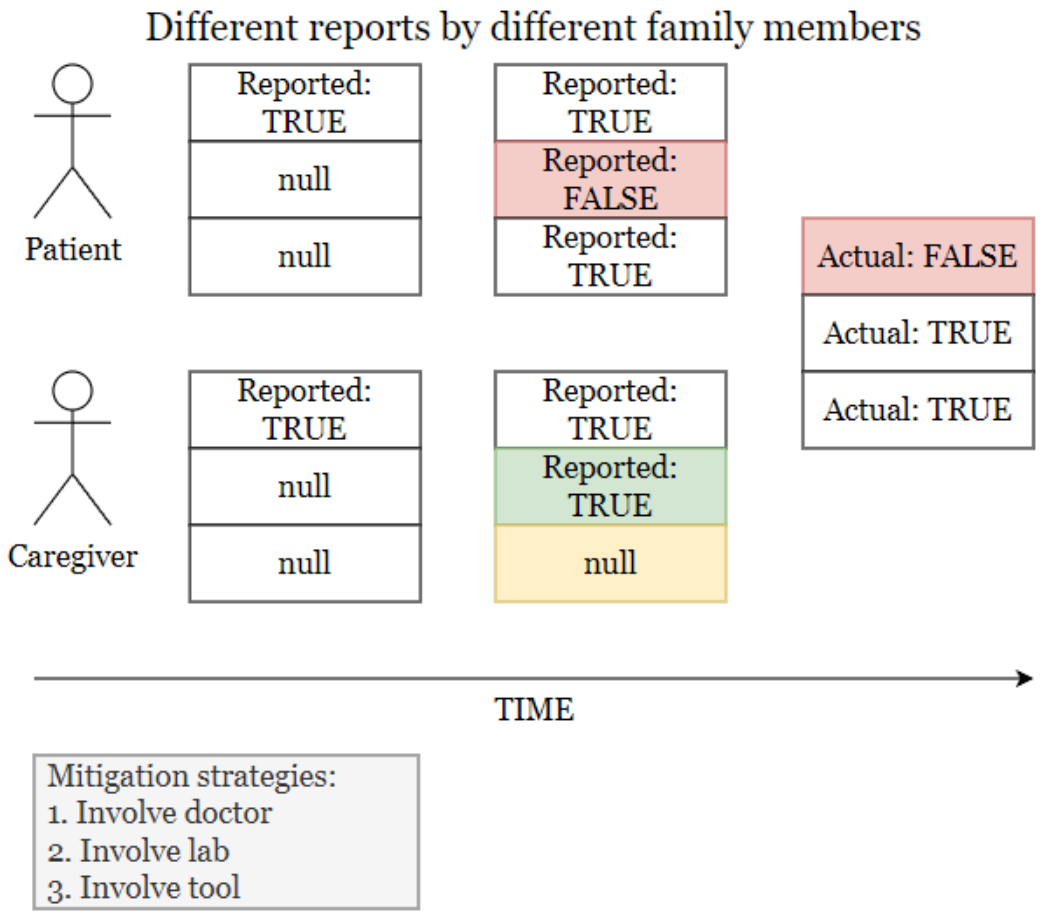



**Fig 6**: The digital tools will uncover false information provided by the patient. The conflicting information can then be used to carry out further investigations.

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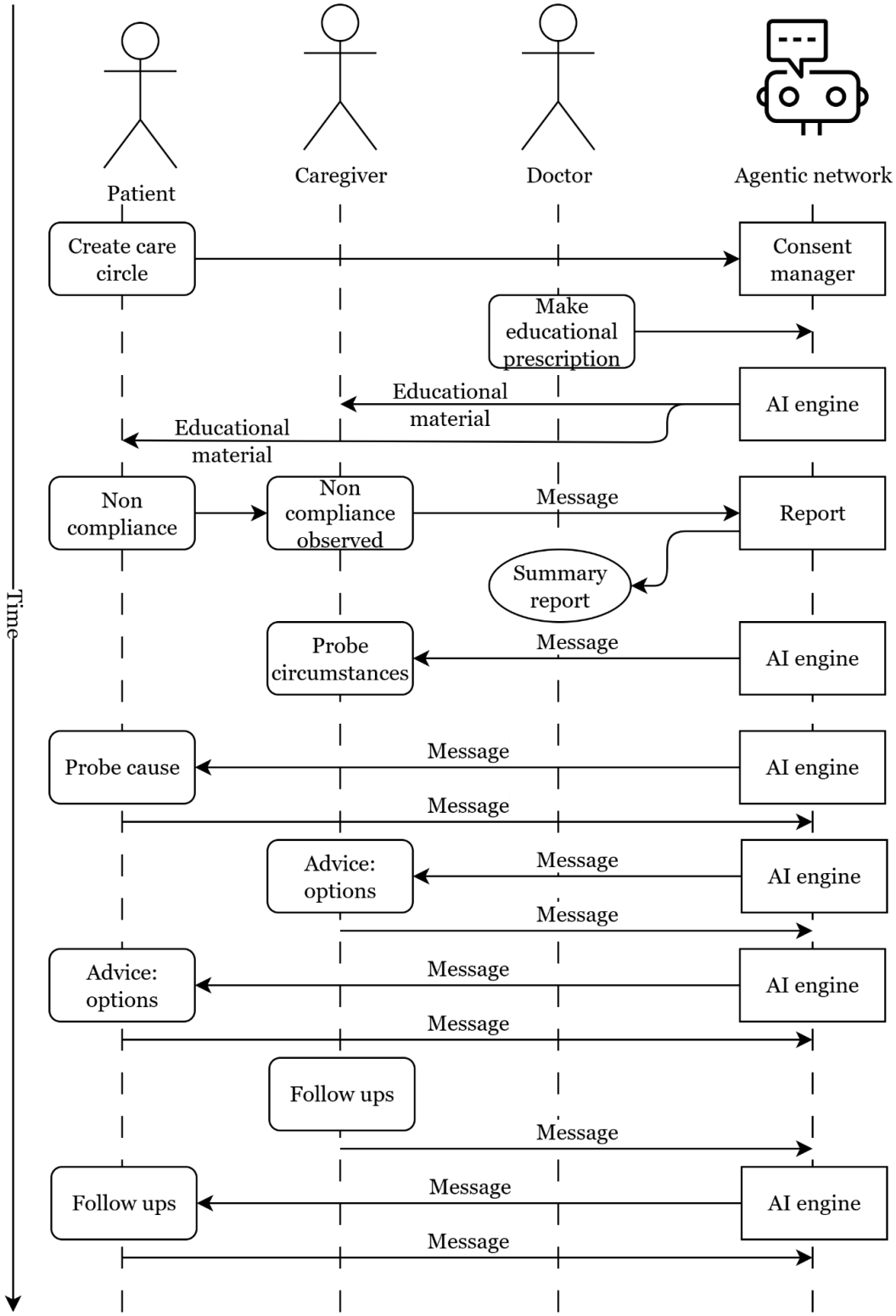



**Fig 7**: The above sequence diagram illustrates **one exemplary operational flow** of the system for a non-adherence identification scenario. The invention is not limited to this specific sequence of interactions or messages

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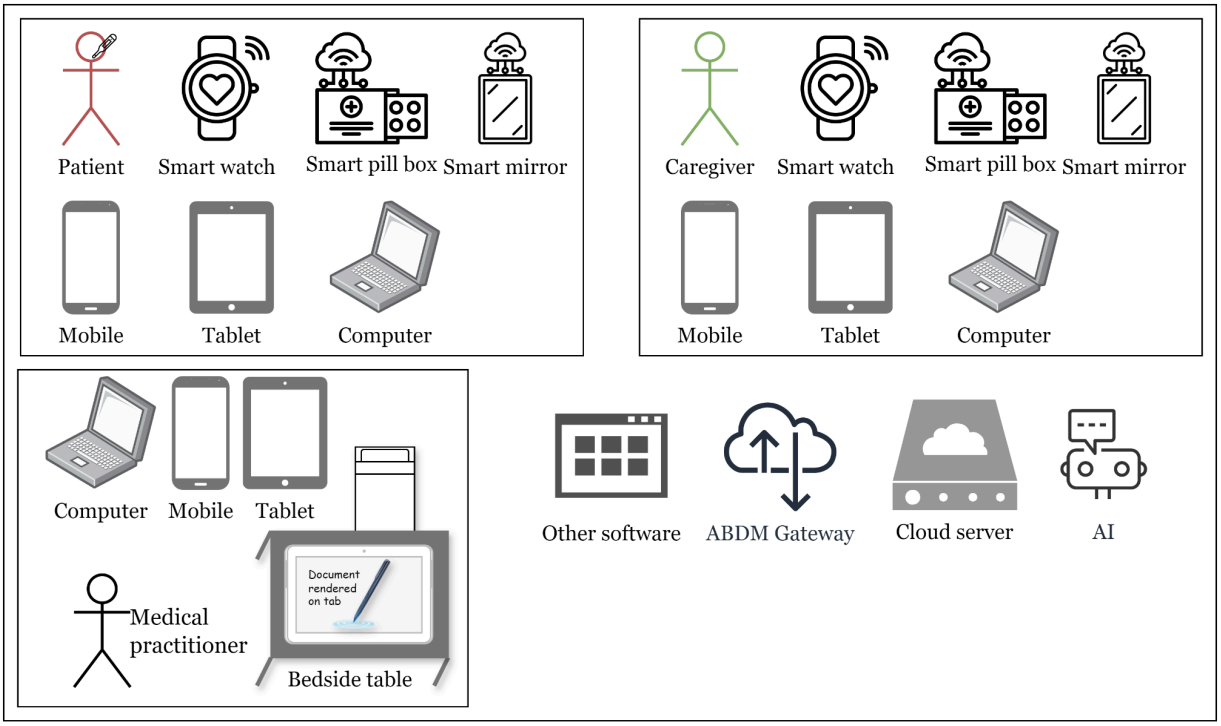
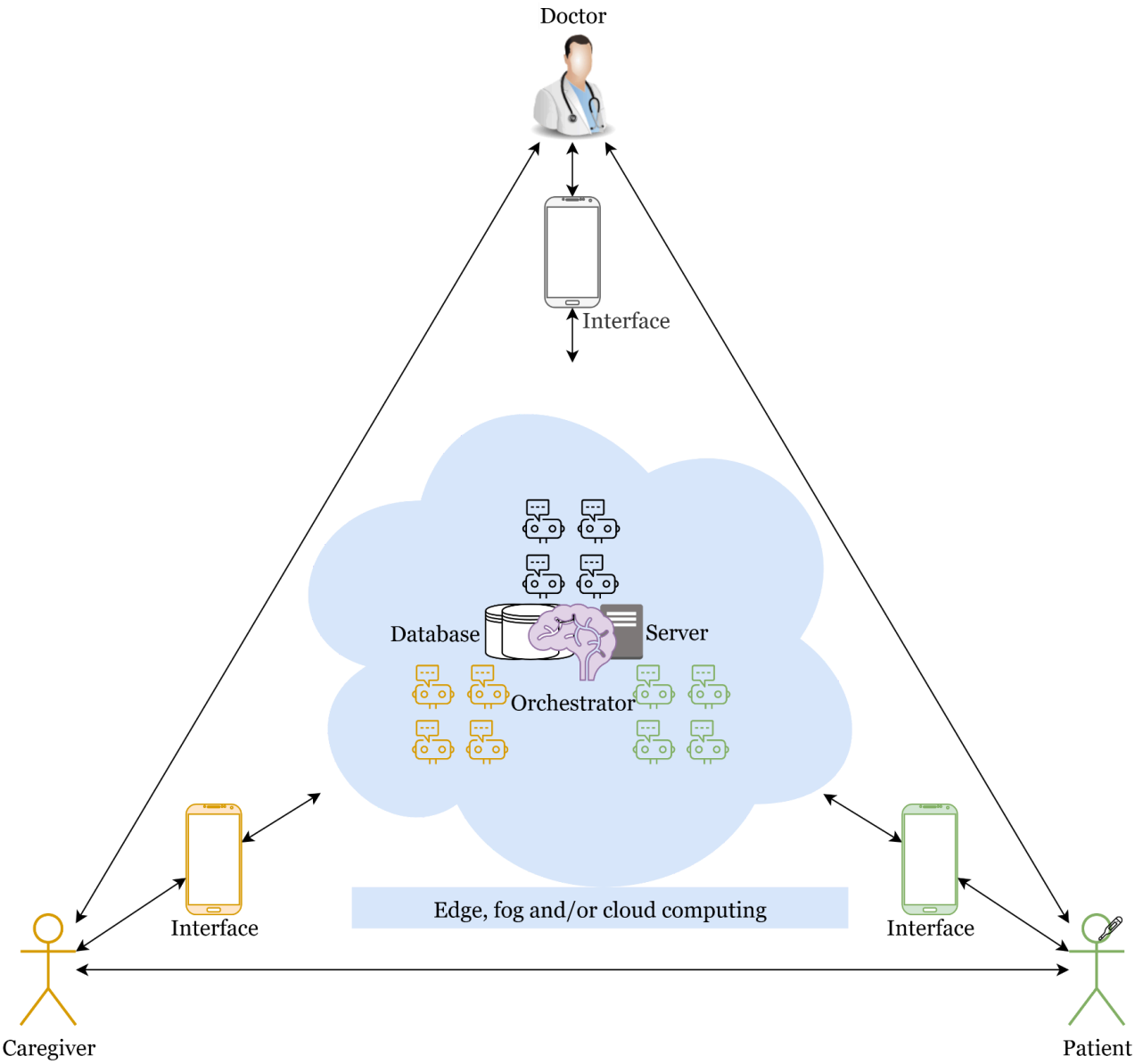



**Fig 8**: Examples of devices used by individuals in the system

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| | |
|---|---|
| Patient's interface includes - 1. Consent manager, 2. Chatbot<br><br>Doctor's interface includes - 1. Dialog graph maker, 2. Conversation analyzer, 3. Direct conversation tool<br><br>Caregiver's interface includes - 1. Discussions about self, 2. Discussions about patient | Most of the computational work, especially related to the orchestrator and optionally related to the agents is done on central or distributed servers.<br><br>Messages may be filtered based on - 1. Consent, 2. Prior availability of information, 3. Manual or auto selection of receivers, 4. Closeness and rapport with patient |

**Fig 9**: An embodiment exemplifying the physical view. Users in a care circle may use personal devices. These may communicate with a cloud server that hosts the orchestrator. Agents may run on client nodes or the server. Example of interfaces used by different users in the system

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Network of agents and humans. Agents communicate information about different individuals in a care circle. Each circle represents a 'private circle.' The agents in each circle focus on the requirements of the user within the circle.

A personal agent alone can leak information about individuals. So a privacy discriminator checks the facts and identifies which informatino should be masked. Safe information is shared by the personal agent directly grey information passes through the orchestrator. The goal based orchestrator continues to watch over the consent hops based on the preferences of each user in a care circle.

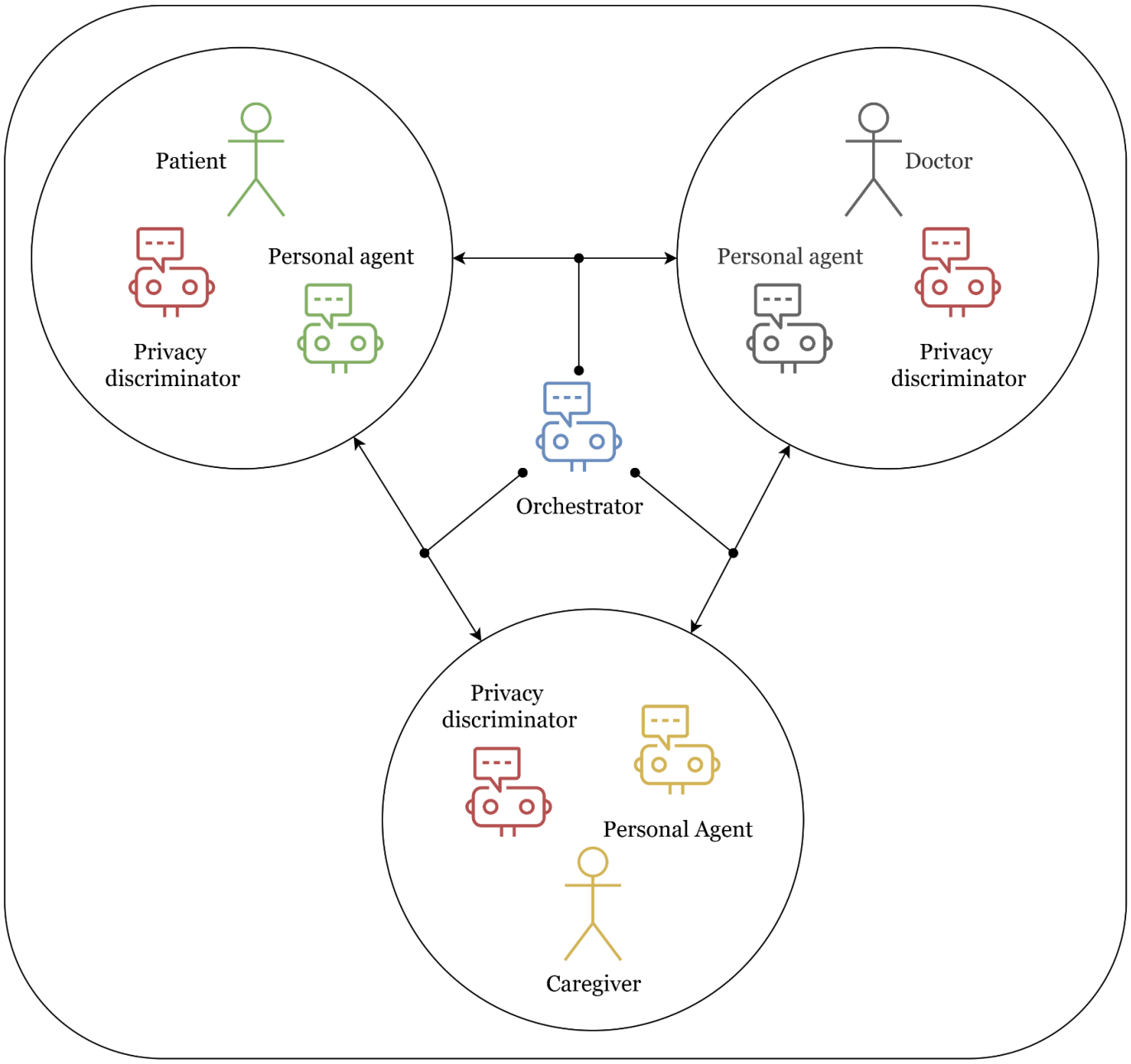



**Fig 10**: The personal assistants create a network of agents. Each circle represents a private and personalized environment. Each private circle thinks selfishly and try to maximize the benefits of the user. The agents may or may not perform computation on a single server. To maintain privacy, each personal agent is monitored by a privacy discriminator. These may run on the user's device or on a centralized server. An - orchestrator will identify when to override a privacy adversary. It will further support in managing consent hops. If a piece of information is restricted between the target and the expert, the need identifier and their agent shall not be able to access it. It will also help personal agents share information with each other effectively.

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Training through adversarial simulations.
For training, the humans are replaced with AI adversarial agents. They simluate real world scenarios like triangulation, breaking the law, hiding information, lying. The judge judges the orchestrator and acts like the 'human in the loop'. This provides the loss function for the training.

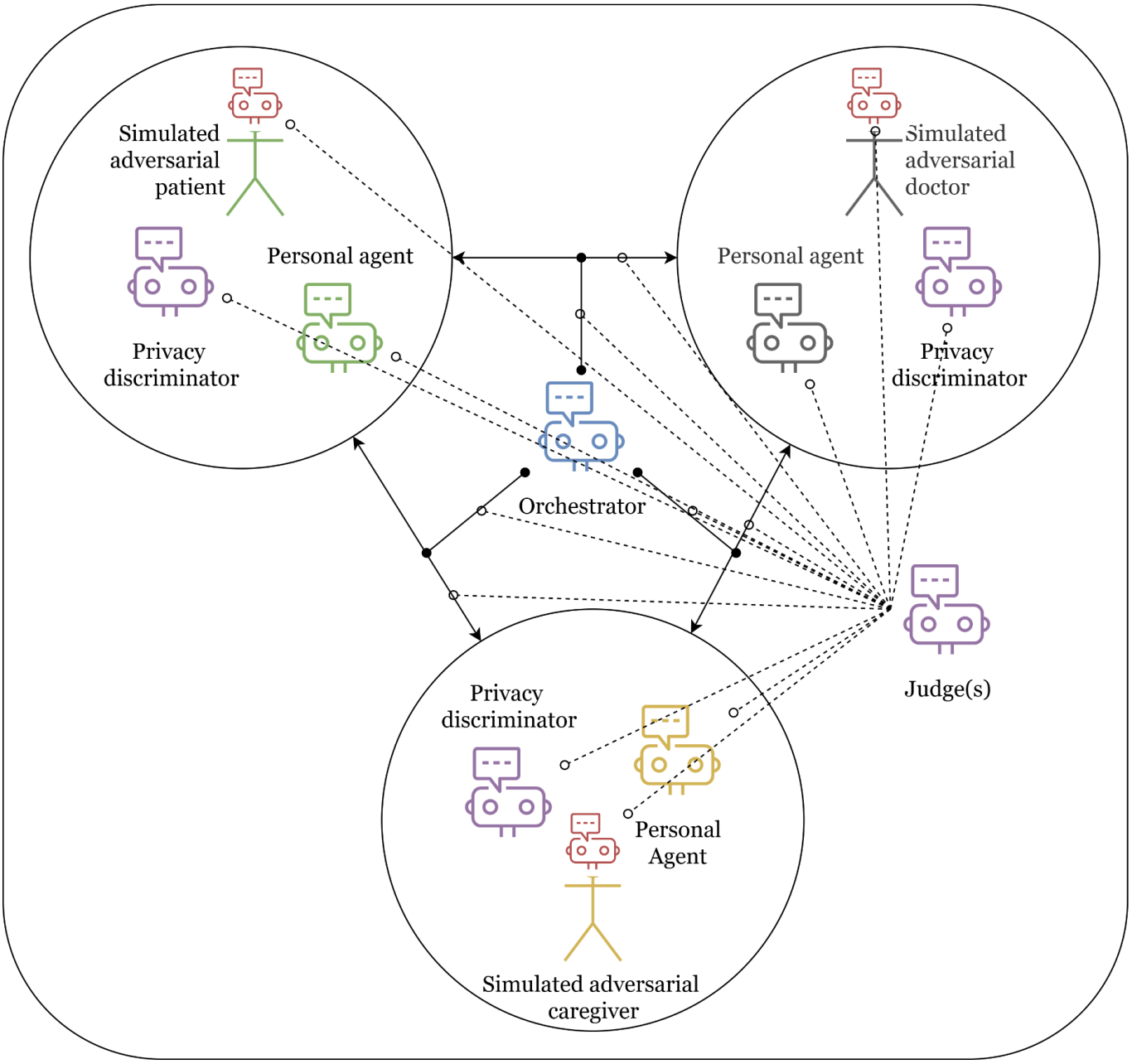



**Fig 11**: For training, the humans are replaced with AI adversarial agents. They simluate real world scenarios like triangulation, breaking the law, hiding information, lying. The judge judges the orchestrator as well as personal/privacy agents and acts like the 'human in the loop'. This provides the rewards and loss function for the training each agent.

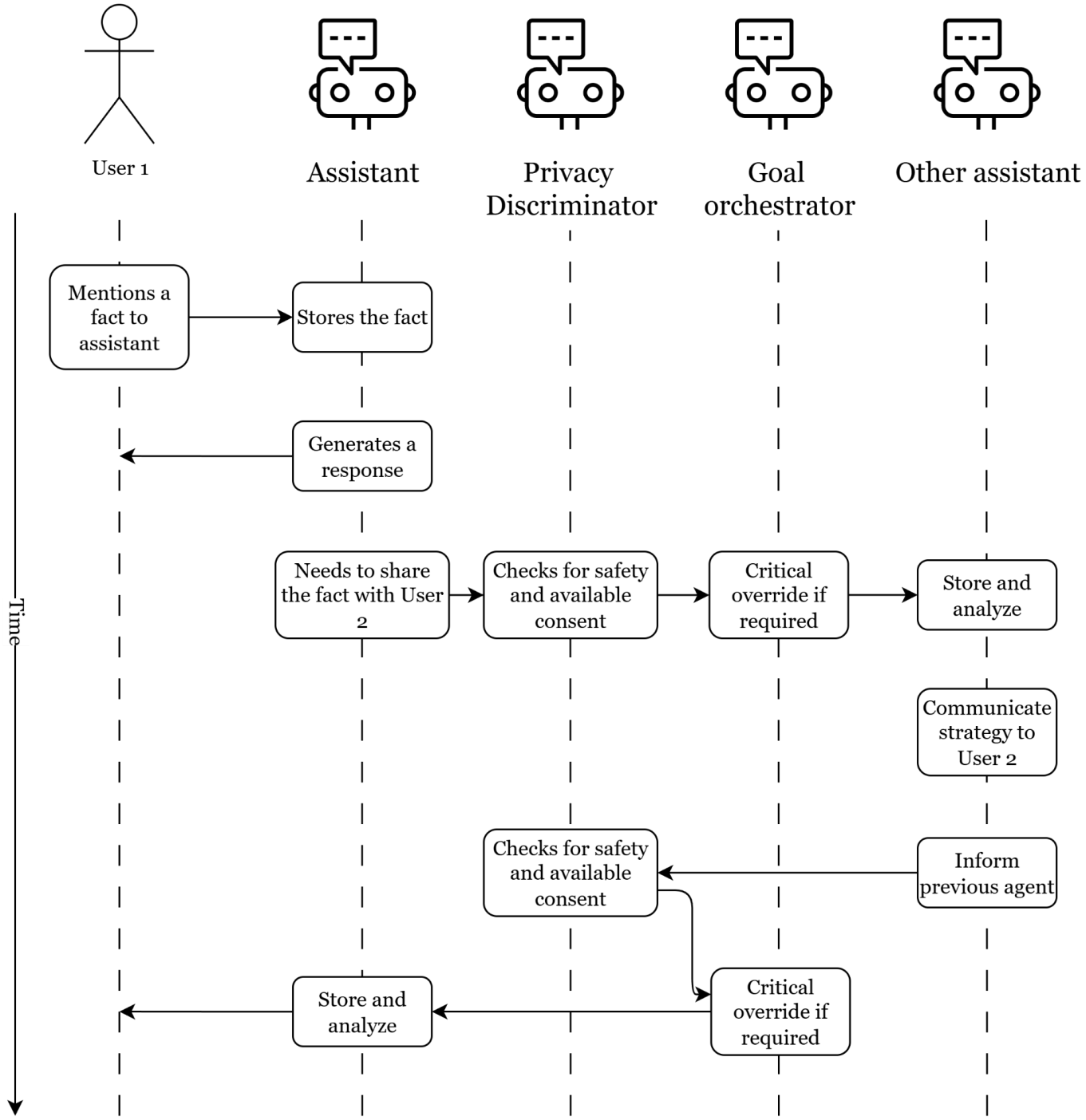



**Fig. 12**: An exemplification of how agents and sub agents may collaborate towards achieving a goal.

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**Fig.13**: Screenshots of the running system running as a command line interface. Clockwise from topleft - Orchestrator server, Patient, Caregiver, Clinician.

# Technical Framework: State Representation & Learning System

- The system's core logic is implemented on a server communicatively coupled to user devices (e.g., smartphones, smart watches). It estimates the true state of compliance ($S_{true}$) by reconciling the patient's report ($R_p$), the caregiver's report ($R_c$), and objective data ($D_{obj}$), weighted by the trustworthiness of each source.
- This estimation is part of a broader learning system designed to evolve and improve over time. The foundation of this system is the **State Vector**.

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## Part 1: The State Vector - Representing the Patient's World

Before we can define how the system learns, we must first define *what* it learns about. The system's understanding of a patient and their care circle at any given moment in time (t) is captured in a high-dimensional, sparse **State Vector**, denoted as S_t. This vector is a structured JSON object that serves as the complete context for all AI decision-making.

The novelty here lies in the multi-faceted nature of the state, which combines clinical, behavioral, social, and conversational data into a single, machine-readable format.

**Structure of the State Vector (S_t)**

The state vector is composed of several key sub-objects. It is inherently sparse; most fields will be `null` at any given time, and the system learns to act on the data that is present.

```
{
  "circle_id": "uuid-for-alice-circle",
  "user_id": "uuid-for-alice",
  "timestamp": 1678886400,
  "clinical_state": {
    "symptoms": {
      "dizziness": { "value": true, "severity": 7, "timestamp": 1678886400 },
      "chest_pain": { "value": true, "severity": 9, "timestamp": 1678886340 }
    },
    "adherence": {
      "medication_X": { "value": "missed_dose", "timestamp": 1678882800 }
    },
    "observations": {
      "blood_pressure": { "value": "150/95", "timestamp": 1678886000 }
    },
    "coded_entries": {
      "snomed_ct": ["29857009"], // Chest Pain
      "icd_10": ["R07.4"]
    }
  },
  "behavioral_state": {
    "stage_of_change": "Contemplation",
    "coping_style": "Denial",
    "irrationality_observed": "Procrastination"
  },
  "social_state": {
    "members": [
      { "user_id": "uuid-for-bob", "role": "caregiver", "relationship": "spouse", "closeness": 85, "consent":
"full" },

      { "user_id": "uuid-for-dr_evans", "role": "doctor", "relationship": "physician", "closeness": 60,
"consent": "full" }
    ],
    "last_interaction_conflict": true
  },
  "conversational_state": {
    "session_state": "AWAITING_CONSENT",
    "pending_action_id": "uuid-for-pending-question",
    "last_agent_action": "REQUEST_CONSENT"
  },
```

```
  "commercial_state": {
    "inquired_about_device": "blood_pressure_monitor",
    "inquired_about_medication": "metformin"
  }
}
```

**Examples and Explanation:**

- **`clinical_state`:** This is the core medical data.
  - **Example:** It logs not just that `chest_pain` is `true`, but also its reported severity and timestamp. This allows for temporal analysis (is the pain getting worse?).
  - **Novelty:** It includes a field for `coded_entries`, where the AI maps natural language to standardized medical ontologies like SNOMED CT and ICD-10, making the data structured and interoperable.
- **`behavioral_state`:** This captures the psychological context using the models we've defined.
  - **Example:** Knowing the patient is in the `"Contemplation"` stage and using `"Denial"` as a coping style allows the Personal Agent to choose a more gentle, motivational interviewing (FRAMES) approach rather than a direct, informational one.
- **`social_state`:** This maps the care circle itself.
  - **Example:** It tracks the `closeness` score and `consent` level for each member, which directly informs the Orchestrator's decisions about information sharing.
- **`conversational_state`:** This is the agent's short-term memory.
  - **Example:** If the `session_state` is `"AWAITING_CONSENT"`, the system knows that the user's next message is likely a response to a question, which prevents the conversational loops you previously identified.

## Part 2: Loss Functions, Rewards, and the Learning Process

With the State Vector (St) defined, we can now architect the mechanisms that drive the system's learning. The training process is bifurcated: we train the **Personal Agents** for conversational quality and the **Orchestrator** for optimal decision-making.

#### 1. Personal Agent Training (Optimizing the Conversation)

The goal of a Personal Agent is to be a trusted, effective conversational partner. Its performance is judged not just on what it says, but on the outcome of its words.

- **Training Method:** Reinforcement Learning with Human Feedback (RLHF). Conversations are logged and later rated by human experts (e.g., clinicians). These ratings provide the reward signal used to fine-tune a base conversational model (like Gemini).
- **Loss Function ($L_{Agent}$):** The agent is trained to minimize a weighted loss function that prioritizes actionability and positive outcomes over generic empathy.
  $L_{Agent} = w_1 L_{Actionability} + w_2 L_{Adherence} + w_3 L_{Conciseness} + w_4 L_{Elicitation} + w_5 L_{Empathy}$
  - $L_{Actionability}$ **(High Weight):** Measures if the agent's reply provides a clear, actionable next step or a goal-driven question.
    - **Example:** A reply like *"Would you like me to inform the doctor?"* after a report of pain receives a higher score than *"That sounds difficult."*
  - $L_{Adherence}$ **(High Weight):** A crucial, delayed reward. If a user later reports completing an action suggested by the agent (e.g., checking their blood pressure), the original message that prompted this action receives a strong positive reward signal.
  - $L_{Conciseness}$ **(Medium-High Weight):** Penalizes verbosity, rewarding the agent for matching the user's conversational style and getting to the point efficiently.
  - $L_{Elicitation}$ **(Medium Weight):** Rewards the agent for replies that successfully elicit new, useful clinical or behavioral facts from the user.

- $L_{Empathy}$ **(Low Weight):** A modulating factor. The response must be appropriate and build rapport, but this is secondary to being effective and actionable.

**2. Orchestrator Training (Optimizing Decisions)**

The Orchestrator is a non-verbal decision engine. It is trained using Reinforcement Learning to select the optimal action based on the current state to maximize long-term "Care Circle Health."

- **State (S):** The comprehensive State Vector, St, as defined in Part 1.
- **Action (A):** A discrete set of choices the Orchestrator can make, including `SHARE`, `ESCALATE`, `REQUEST_CLARIFICATION`, `REQUEST_CONSENT`, and `WAIT`.
- **Reward (R(t)):** The core of the learning process. After an action is taken, a human reviewer provides a score based on its outcome. The full reward structure is extensive:

**2.1. Orchestrator Reward Function (R(t)):**

This function trains the central decision-making engine. The model is rewarded or penalized based on the quality of its actions, as determined by a human clinical reviewer.

- **Safety & Clinical Outcomes**
  - Timely Critical Escalation (e.g., doctor confirms emergency): **+100**
  - Missed Critical Escalation: **-200**
- **Data Quality & Accuracy**
  - Successful Conflict Resolution (via mediated clarification): **+20**
  - Correctly mapping user's language to a medical code (e.g., ICD-10): **+30**
- **Health Outcomes & Adherence**
  - Observing a user follow through on advice (from AI, doctor, or caregiver): **+50**
- **User Experience**
  - A successful, consented share of information that leads to a positive interaction: **+5**
  - An annoying or redundant request for consent: **-10**
- **Privacy & Security**
  - Sharing sensitive information that violates the user's consent level: **-100**

**3. Training with Adversarial Simulations**

To ensure the system is robust, safe, and secure, a significant part of the training involves adversarial simulations. The system is trained not just on ideal scenarios but on deliberately challenging ones designed to test its limits.

- **The Goal:** To train the system to be resilient against misinformation and to handle complex social dynamics.
- **Method:** We create "synthetic adversarial users"—AI agents programmed with specific, non-cooperative goals. The Orchestrator and Personal Agents are then trained to successfully navigate these scenarios.
- **Adversarial Scenarios Include:**
  - **Triangulation:** A scenario with three or more members providing conflicting reports on the same fact (e.g., Patient says yes, Caregiver 1 says no, Caregiver 2 says maybe). This trains the conflict resolution engine.
  - **Secrets and Lies:** A synthetic user is programmed to consistently hide information or provide false reports (e.g., a patient persona in "Denial" who always claims to have taken their medication). The system is rewarded for identifying long-term inconsistencies between this user's reports and those of other members.
  - **Financial Adversaries:** A persona is designed to manipulate the system for a non-health outcome (e.g., exaggerating symptoms to get a work excuse). The Orchestrator is rewarded for prioritizing objective facts and flagging subjective, uncorroborated claims to the doctor.

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# Chain of Thought and Agentic Interaction

This document provides a comprehensive overview of the multi-agent system's interaction and thought process, focusing on the present implementation and clear agent responsibilities.

## 1. System Overview

The multi-agent healthcare system uses a sophisticated chain of thought approach where multiple specialized agents collaborate to process user messages, make decisions, and coordinate actions. Each agent has distinct responsibilities and operates within a well-defined framework.

## 1.1 Component Definitions

Before diving into the architecture, it's crucial to understand the different types of components in our system:

**Agents** 🤖

- **Purpose**: AI-powered conversational entities that interact with users
- **Examples**: PersonalAgent, CareAgent, DoctorAgent
- **Function**: Generate natural language responses, build rapport, elicit information
- **Location**: `agents/` directory

**Handlers** ⚙️

- **Purpose**: Business logic modules that process specific types of requests
- **Examples**: SharingHandler, ConflictHandler, ActionExecutor
- **Function**: Execute rules, make decisions, coordinate actions
- **Location**: `handlers/` directory

**Prompts** 📝

- **Purpose**: Templates that guide AI agent behavior and responses
- **Examples**: unified_prompt.js, doctor_prompts.js
- **Function**: Define conversation style, safety rules, response patterns
- **Location**: `prompts/` directory

**Modules** 📦

- **Purpose**: Reusable code components that provide specific functionality
- **Examples**: Database queries, utility functions, state management
- **Function**: Data access, calculations, state tracking
- **Location**: Various directories (`db/`, `utils/`, etc.)

**Key Distinction**: Agents are the "personalities" that talk to users, Handlers are the "brains" that make decisions, Prompts are the "instructions" that guide agent behavior, and Modules are the "tools" that provide functionality.

## 2. Agent Architecture

## 2.1 Agent Types & Responsibilities

**PersonalAgent**

- **Purpose**: Serves as the primary conversational interface for the patient
- **Context**: Active when the user is the primary patient in their own care circle

- **Responsibilities**:
    - Builds trust and rapport with the patient
    - Elicits health information and concerns
    - Provides empathetic responses
    - Facilitates self-reporting of symptoms and behaviors

**CareAgent**

- **Purpose**: Serves as the conversational interface for caregivers and family members
- **Context**: Active when the user is in someone else's care circle (as a caregiver)
- **Responsibilities**:
    - Provides context-aware responses about the patient's health
    - Facilitates caregiver-patient communication
    - Offers support and guidance to caregivers
    - Mediates sensitive conversations

**DoctorAgent**

- **Purpose**: Specialized agent for medical professionals
- **Context**: Active when a doctor is interacting with the system
- **Responsibilities**:
    - Provides clinical context and medical insights
    - Handles escalation management
    - Facilitates doctor-patient communication
    - Manages medical decision-making processes

**Handlers**

- Each agent is made up of multiple handlers. Each handler is an individual model and responsible for effective execution of a single task.
- Handlers can be shared among agents due to the reciprocity of role in care circles.

## 2.2 Agent Identity Management

The system dynamically determines which agent to use based on the user's context:

```
None
// Agent Identity Logic
async function getAgentName(userId, circleId) {
    const currentCircle = await findCircleByPatientId(circleId);
     return userId === currentCircle.primary_patient_id ? 'PersonalAgent'
: 'CareAgent';
}
```

## 2.3 Reciprocity of agents

This section explains how users and agents adapt their roles based on context, creating fluid care circles where individuals can be both patients and caregivers.

The roles of caregiver, patient and doctor have reciprocity. A doctor may fall ill and become a patient. A caregiver may fall ill and become a patient, a patient may act as a caregiver when the spouse falls ill. Similarly, the personal agents may switch their roles if required. This may be done using explicit commands like ‘/switch,’ intended to change the use case or care circle. It may be done using implied methods in which the user only mentions the issues faced by the user or a member of the care circle. An example can be couple and family

counselling in which the initial patient (say husband) discloses a behavioural problem of the spouse, and hence becomes a caregiver for the wife who now holds the patient role.

### 2.3.1 Role Reciprocity Framework

**Role Reciprocity Explained:**

- **Same Person, Different Roles**: A user can be a patient in their own circle and a caregiver in their spouse's circle
- **Agent Adaptation**: The same user gets PersonalAgent (patient) or CareAgent (caregiver) based on context
- **Dynamic Switching**: Users can switch between circles using `/switch` commands
- **Context-Aware Responses**: Agents adapt their language and capabilities based on the user's role

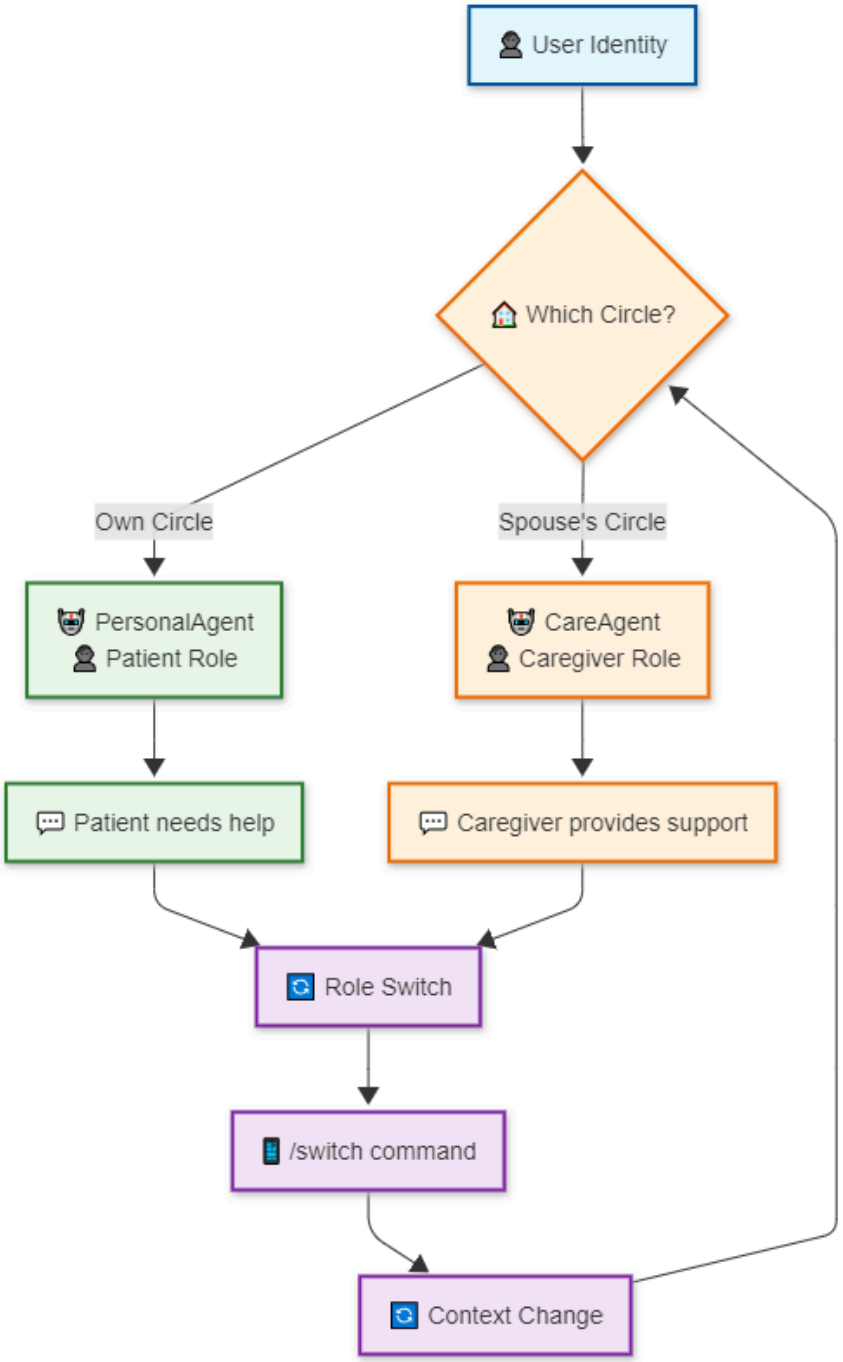



**Fig 14**: This diagram shows how users and agents dynamically adapt their roles based on the care circle context:

### 2.3.2 Care Circle Fluidity

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**Care Circle Fluidity Explained:**

- **Two Patients**: Patient A and Patient B each have their own care circles
- **Spouse Caregivers**: Spouse A cares for Patient B, Spouse B cares for Patient A
- **Sibling Caregivers**: Sibling C provides care in both circles
- **Role Switching**: Spouses can switch between patient and caregiver roles
- **Shared Caregivers**: Siblings can be caregivers in multiple circles

This framework provides a robust foundation for the multi-agent healthcare system, ensuring clear responsibilities, effective communication, and reliable decision-making processes.

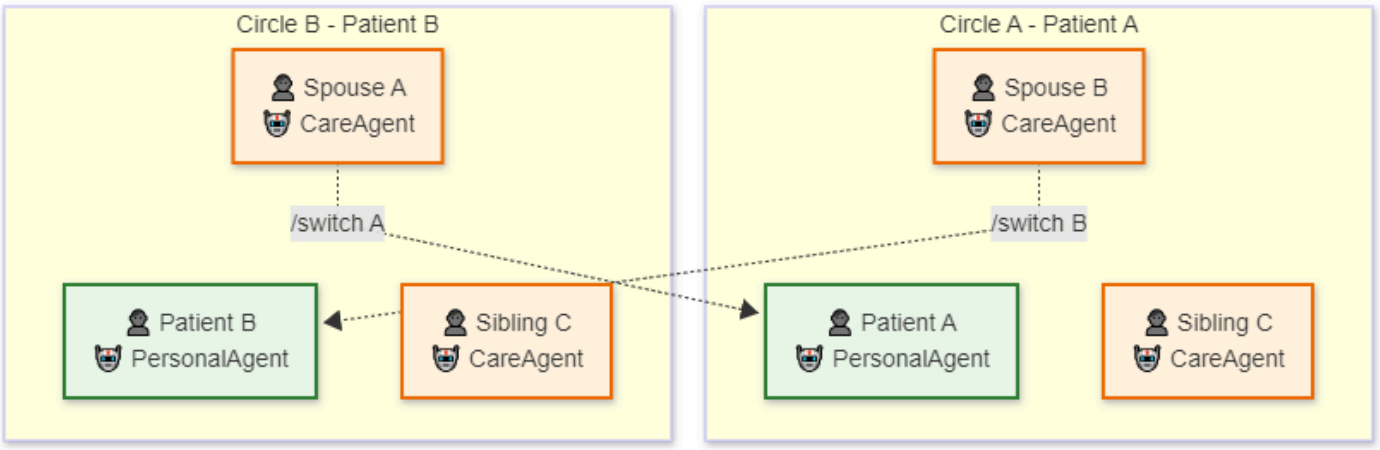



**Fig 15:** This diagram shows how care circles overlap with spouses and siblings as caregivers

# 3. Detailed Architecture of the Core Logic Engine

The novelty of the invention is not merely in the connection of agents, but in the sophisticated, multi-step "chain of thought" process that the system's core logic engine (the 'Orchestrator' and its associated handlers) executes for each message. This process is what allows the system to manage complex conversational state, resolve conflicts, and make safe, context-aware decisions.

The diagrams in the following sections provide a detailed disclosure of this novel internal architecture.

## 3.1 Main Processing Flow: The "Chain of Thought"

The flowchart below illustrates the main "chain of thought" for the core logic engine. This is the central process that is initiated every time a user sends a message. It shows how the system dispatches the message to a series of specialized sub-systems (the colored boxes) in a specific, ordered sequence to ensure the correct action is always taken.

The following flowchart provides a detailed view of the system's 'chain of thought' for **one preferred embodiment** of the invention. This illustrates how the various specialized handlers and agents collaborate to process a user's message. This diagram shows the chain of thought process from when a user sends a message to when they receive a response. Each color represents a different type of componen. Specifics of the steps are highlighted in subsequent diagrams.

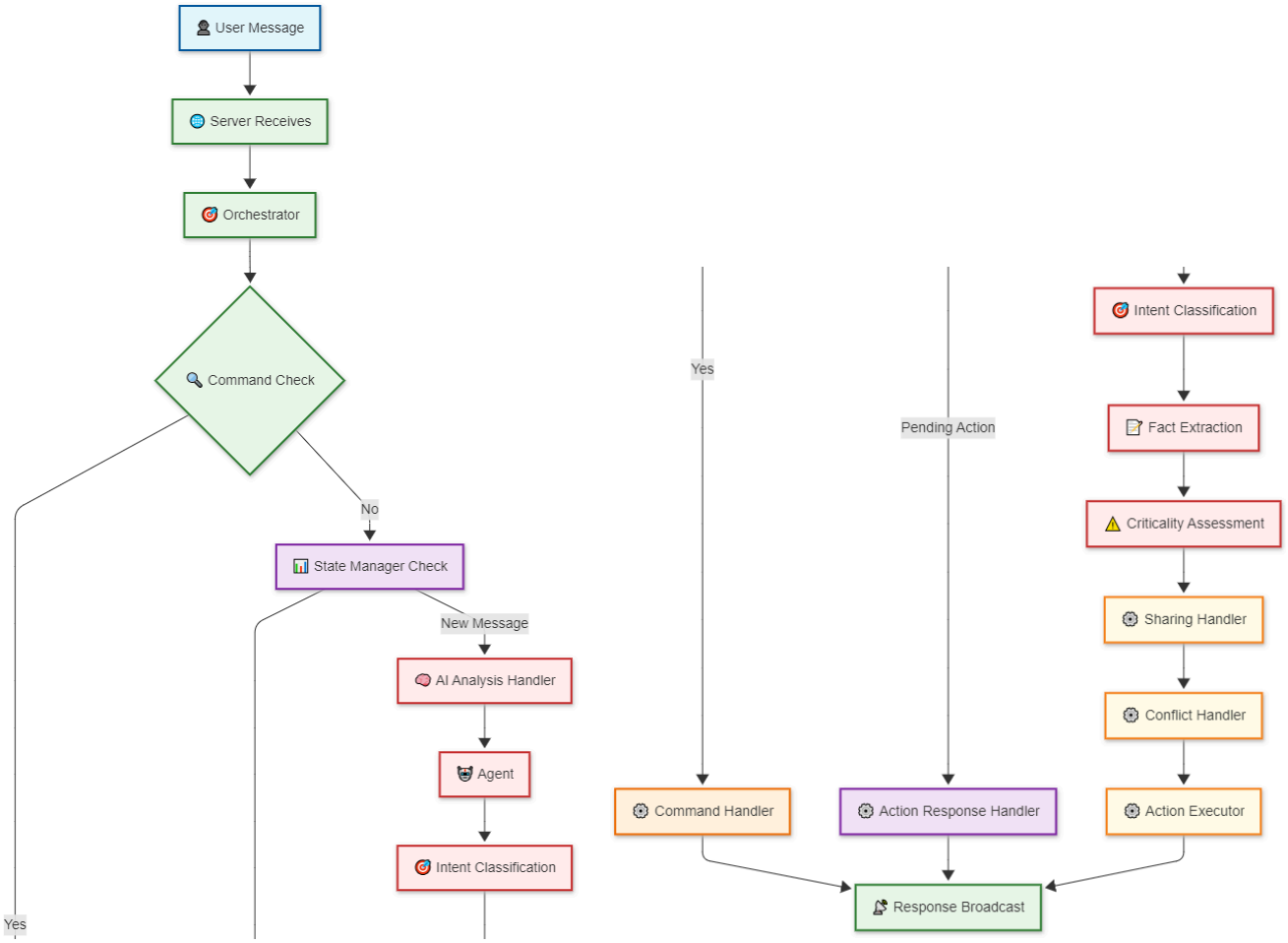



**Fig. 16**: Main Message Processing Flow. This diagram shows the complete chain of thought process from when a user sends a message to when they receive a response. Each color represents a different type of component:

**Legend:**

- 🔵 **User Input**: Messages from users
- 🟢 **Orchestration**: Core workflow management
- 🟠 **Command Processing**: Direct user commands
- 🟣 **State Management**: Conversation state tracking
- 🔴 **AI Processing**: LLM analysis and decision making
- 🟡 **Business Logic**: Specialized handlers for different scenarios
- 🟢 **Output**: Response delivery to users

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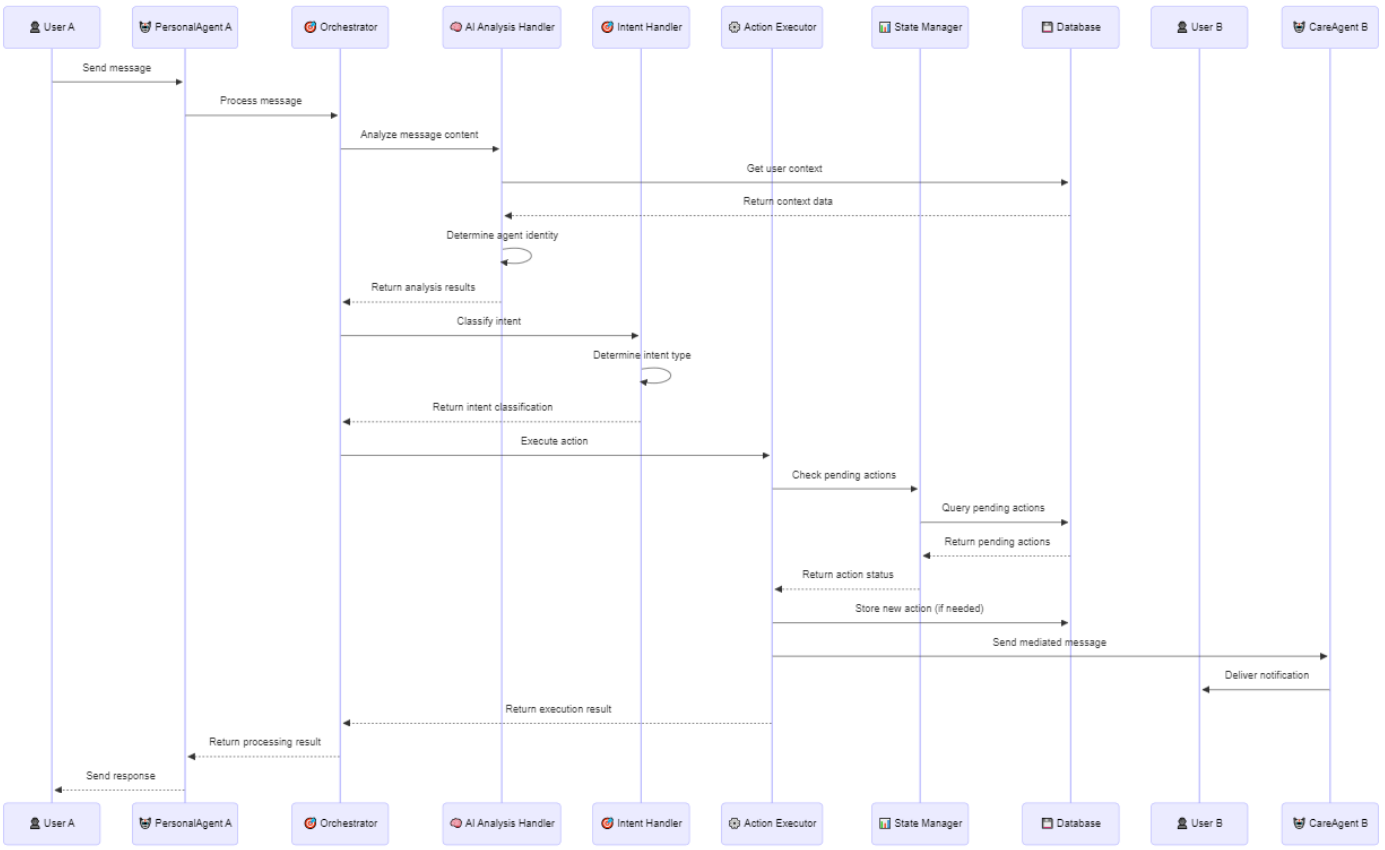



**Fig. 17**: This diagram shows the basic message flow when a user sends a message to their agent and the system processes it through the orchestrator to determine the appropriate response.

## Key Interactions

### 1. Message Reception

- **User A** sends a message to their **PersonalAgent A**
- The agent forwards the message to the **Orchestrator** for processing

### 2. AI Analysis Phase

- **Orchestrator** delegates to **AI Analysis Handler**
- Handler retrieves user context from database
- Determines agent identity (PersonalAgent vs CareAgent)
- Returns analysis results to orchestrator

### 3. Intent Classification

- **Orchestrator** uses **Intent Handler** to classify the message intent
- Intent types include: fact reporting, doctor requests, mediation, advice sharing
- Returns intent classification to orchestrator

### 4. Action Execution

- **Orchestrator** delegates to **Action Executor**
- **Action Executor** checks for pending actions via **State Manager**
- **State Manager** queries database for pending actions
- New actions are stored in database if needed

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### 5. Mediated Communication

- **Action Executor** sends mediated message to **CareAgent B**
- **CareAgent B** delivers notification to **User B**
- **Action Executor** returns execution result to orchestrator

### 6. Response Delivery

- **Orchestrator** returns processing result to **PersonalAgent A**
- **PersonalAgent A** sends response back to **User A**

## Component Roles

- **User A**: Patient or caregiver sending the initial message
- **PersonalAgent A**: User A's personal AI agent
- **Orchestrator**: Central decision-making component
- **AI Analysis Handler**: Analyzes message content and context
- **Intent Handler**: Classifies message intent
- **Action Executor**: Executes the determined action
- **State Manager**: Manages pending actions and state
- **Database**: Stores user context, pending actions, and system state
- **User B**: Target user receiving mediated information
- **CareAgent B**: User B's care agent (when User B is in caregiver role)

## Data Flow

1. **Message**: User input with context
2. **Analysis**: AI analysis results with agent identity
3. **Intent**: Classified intent type
4. **Action**: Executed action with state updates
5. **Mediation**: Mediated message to target user
6. **Response**: Final response to originating user

# 3.2 AI Analysis Chain

This diagram details the AI processing steps that occur within the "AI Analysis Handler" box from the main flow diagram above. It shows how the Unified Agent processes messages through multiple analysis stages:

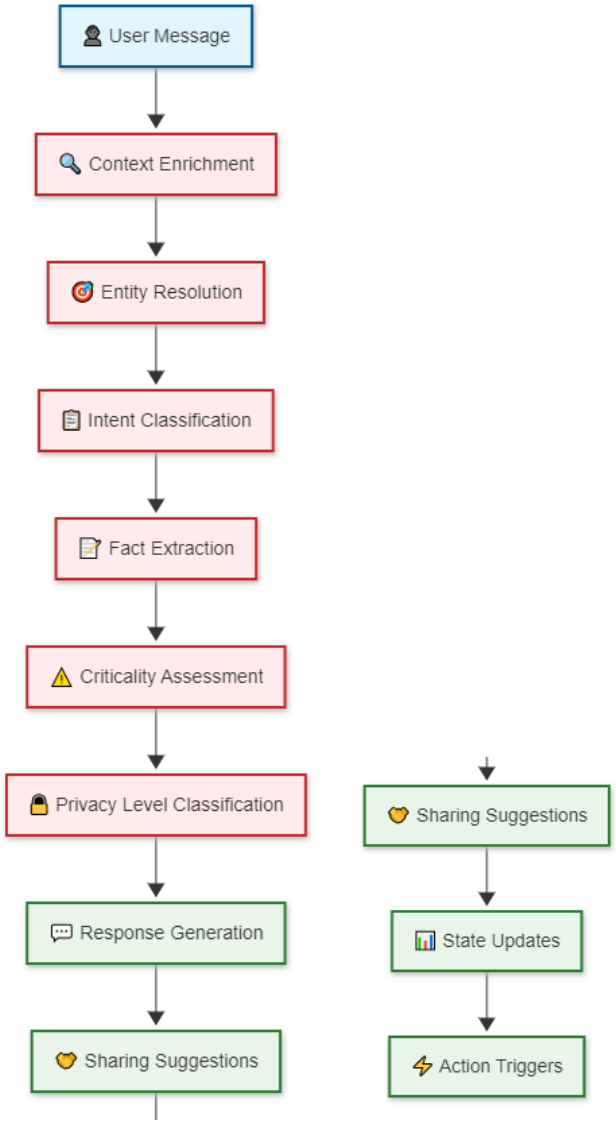



**Fig. 18**: Detailed View of the AI Analysis Sub-System

**This section explains the AI Analysis Chain** (referring to the red boxes in the main diagram):

- **Context Enrichment**: Gathers user history, care circle context, and behavioral models
- **Entity Resolution**: Identifies people, places, and medical terms mentioned
- **Intent Classification**: Determines what the user wants (question, statement, command)
- **Fact Extraction**: Pulls out factual information about health, behavior, or events
- **Criticality Assessment**: Evaluates urgency (LOW, MEDIUM, HIGH)
- **Privacy Level Classification**: Determines sensitivity (PUBLIC, PRIVATE, SENSITIVE)
- **Response Generation**: Creates the agent's conversational reply
- **Sharing Suggestions**: Identifies information that should be shared with others
- **State Updates**: Modifies user's behavioral state and preferences
- **Action Triggers**: Initiates specific system actions (escalation, consent requests)

# 4. Specialized Handlers

**This section explains the Business Logic components** (referring to the yellow boxes in the main diagram). These handlers implement the decision-making rules that determine how information flows between users.

## 4.1 Sharing Handler Package

The sharing logic has been modularized into specialized handlers. This diagram shows how the "Sharing Handler" from the main flow branches into different specialized processors:

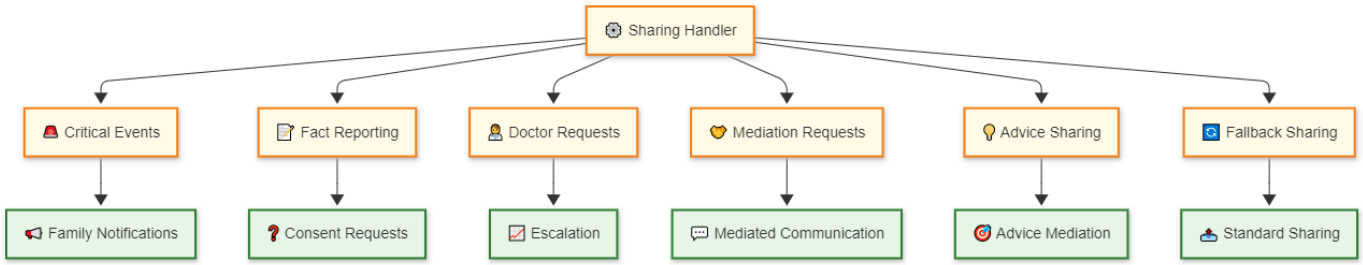



**Fig 19**: Detailed View of the Sharing Handler Sub-System

**Why this section exists**: The main diagram shows "Sharing Handler" as a single yellow box, but this section breaks down what actually happens inside that box. Each specialized handler processes different types of sharing scenarios.

**Handler Responsibilities**

**Critical Events Handler**:

- Detects HIGH criticality events
- Triggers immediate doctor escalation
- Notifies family members of urgent health concerns
- Provides confirmation to the reporting user

**Fact Reporting Handler**:

- Processes facts about other people in the care circle
- Determines if escalation or consent is needed
- Handles cross-circle fact reporting
- Manages consent workflows

**Doctor Requests Handler**:

- Handles explicit requests to speak with doctors
- Creates doctor consultation escalations
- Provides confirmation of doctor notification

**Mediation Requests Handler**:

- Facilitates communication between care circle members
- Generates mediated messages
- Confirms mediation initiation

**Advice Sharing Handler**:

- Processes advice given by one user for another
- Mediates advice through the target user's agent
- Ensures appropriate context delivery

**Fallback Sharing Handler**:

- Handles standard sharing suggestions
- Manages consent-based information sharing
- Provides contextual messages to target users

# 4.2 Conflict Detection & Resolution

This section details the specific conflict resolution process that occurs when users report contradictory information. The main diagram shows "Conflict Handler" as a yellow box, but this section explains what happens inside that box.

**How Conflict Detection Works:**

1. **Conflict Check**: When a new fact is reported, the system compares it against existing facts about the same topic from other users
2. **Conflict Detection**: If contradictory information exists (e.g., Patient says "took medication", Caregiver says "didn't take medication"), a conflict is flagged
3. **Mediated Question Generation**: The system generates a neutral, clarifying question to help resolve the conflict

**What is a Mediated Question?** A mediated question is a carefully crafted, neutral question that the AI agent asks to help resolve conflicting information. It's designed to:

- Avoid taking sides or showing bias
- Ask for specific, factual clarification
- Use gentle, non-confrontational language
- Focus on observable facts rather than opinions

**Example Mediated Questions:**

- "I'd like to clarify about the medication - can you tell me what you observed about [patient's] medication routine today?"
- "There seems to be different information about [patient's] symptoms. Could you describe what you noticed about their condition this morning?"
- "I want to make sure I understand correctly - when did you last see [patient] take their medication?"

**Clarification Generation Process:** The system uses the `generateMediatedClarification` function to create these questions by:

1. Analyzing the conflicting facts
2. Identifying the specific discrepancy
3. Generating a neutral question that asks for clarification
4. Ensuring the question doesn't reveal who reported what conflicting information

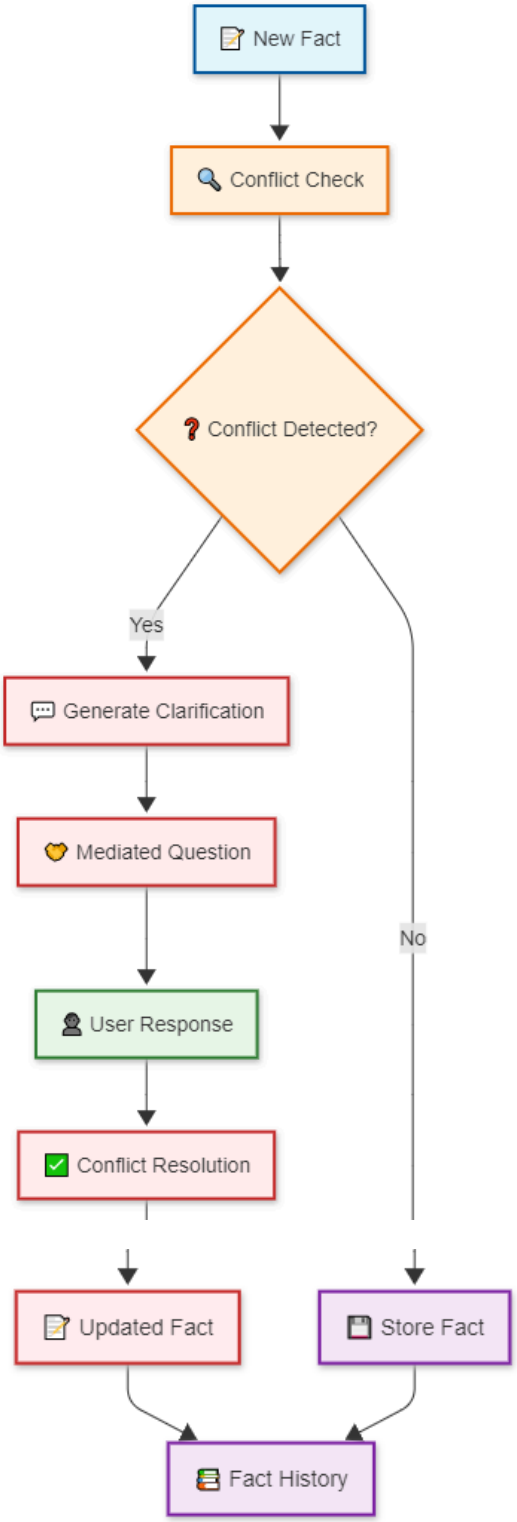



**Fig 20: Conflict Detection Flowchart:** A flowchart detailing the conflict resolution sub-system, from conflict detection to the generation and resolution of a mediated question.

**Fact Update Behavior**: When facts are updated, the system **creates a log entry** rather than overwriting. This maintains a complete history of all reported information, allowing the system to track how facts change over time and identify patterns in conflicting reports.

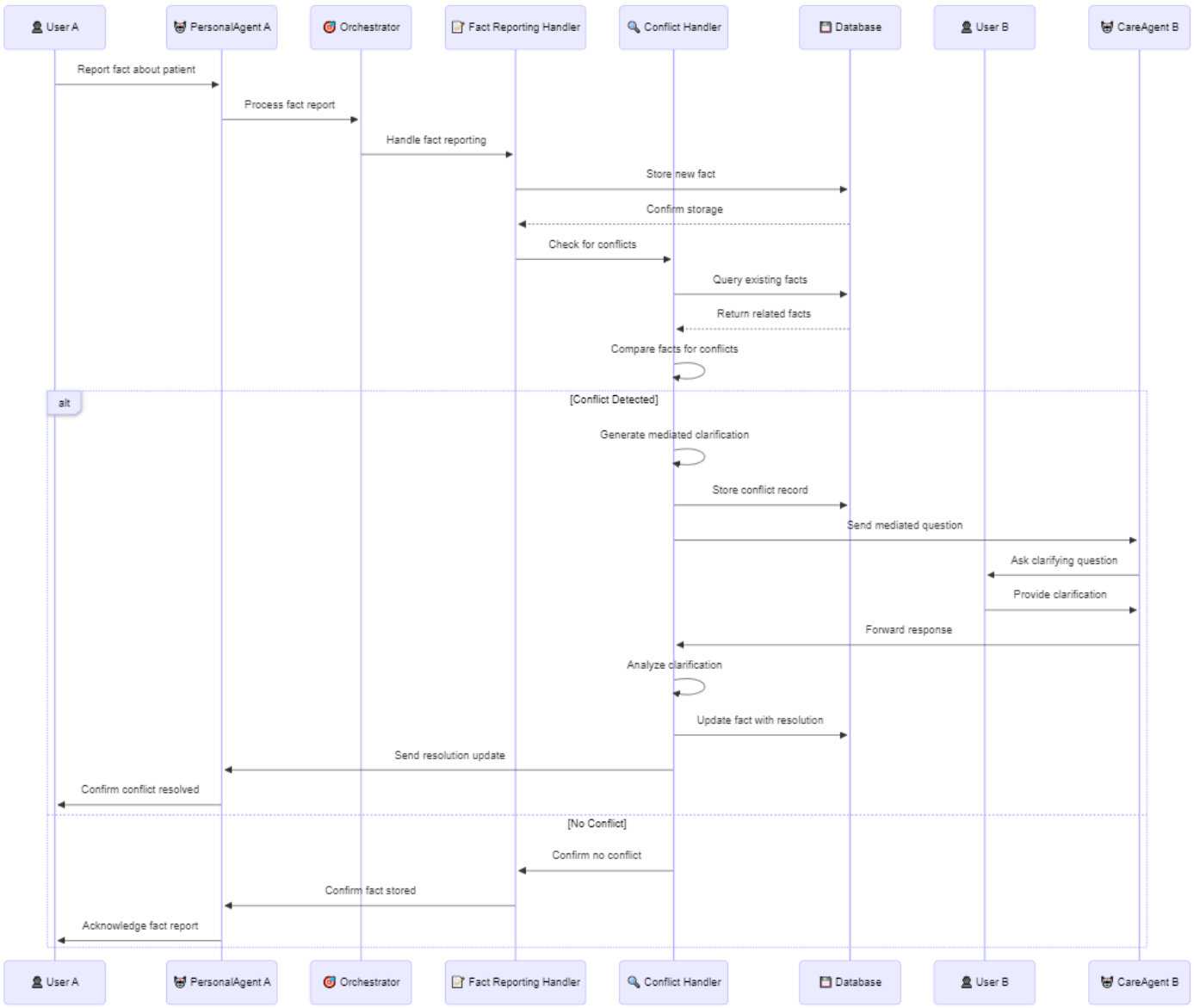



**Fig 21**: This diagram shows the conflict detection and resolution process when users report contradictory information about the same topic.

## Key Interactions

### 1. Fact Reporting

- **User A** reports a fact about the patient to their **PersonalAgent A**
- The agent forwards the fact to the **Orchestrator** for processing
- **Orchestrator** delegates to **Fact Reporting Handler**

### 2. Fact Storage

- **Fact Reporting Handler** stores the new fact in the database
- Database confirms successful storage
- Handler then checks for potential conflicts

### 3. Conflict Detection

- **Fact Reporting Handler** delegates to **Conflict Handler**
- **Conflict Handler** queries database for existing facts about the same topic

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- Database returns related facts for comparison
- **Conflict Handler** compares the new fact against existing facts

### 4. Conflict Resolution Path (if conflict detected)

- **Conflict Handler** generates a mediated clarification question
- Stores conflict record in database
- Sends mediated question to **CareAgent B** (User B's agent)
- **CareAgent B** delivers the clarifying question to **User B**

### 5. User Response and Resolution

- **User B** provides clarification to their agent
- **CareAgent B** forwards the response to **Conflict Handler**
- **Conflict Handler** analyzes the clarification
- Updates the fact in database with resolution
- Sends resolution update to **PersonalAgent A**
- **PersonalAgent A** confirms conflict resolution to **User A**

### 6. No Conflict Path

- If no conflict is detected, **Conflict Handler** confirms no conflict
- **Fact Reporting Handler** confirms fact storage to **PersonalAgent A**
- **PersonalAgent A** acknowledges the fact report to **User A**

## Component Roles

- **User A**: User reporting the initial fact
- **PersonalAgent A**: User A's personal AI agent
- **Orchestrator**: Central decision-making component
- **Fact Reporting Handler**: Handles fact storage and conflict checking
- **Conflict Handler**: Detects conflicts and manages resolution process
- **Database**: Stores facts, conflict records, and resolution updates
- **User B**: User who may have conflicting information
- **CareAgent B**: User B's care agent (when User B is in caregiver role)

## Conflict Detection Logic

### Fact Comparison

- **Topic Matching**: Facts about the same subject (medication, symptoms, behavior)
- **Time Proximity**: Facts reported within a reasonable time frame
- **Contradiction Analysis**: Direct contradictions (took/didn't take medication)
- **Context Consideration**: Similar situations with different outcomes

### Mediated Question Generation

- **Neutral Language**: Avoids taking sides or showing bias
- **Specific Focus**: Asks for clarification on the specific discrepancy
- **Observable Facts**: Focuses on what can be observed rather than opinions
- **Privacy Protection**: Doesn't reveal who reported what conflicting information

## Data Flow

1. **Fact Report**: Initial fact from User A

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2. **Fact Storage**: New fact stored in database
3. **Conflict Check**: Comparison with existing facts
4. **Mediated Question**: Neutral clarifying question (if conflict detected)
5. **User Response**: Clarification from User B
6. **Resolution**: Updated fact with conflict resolution
7. **Confirmation**: Resolution confirmation to User A

### Example Scenarios

#### Medication Conflict

- **User A**: "Patient took medication at 8 AM"
- **User B**: "Patient didn't take medication this morning"
- **Mediated Question**: "I'd like to clarify about the medication - can you tell me what you observed about the patient's medication routine this morning?"

#### Symptom Conflict

- **User A**: "Patient is feeling better today"
- **User B**: "Patient seems worse, more tired than usual"
- **Mediated Question**: "There seems to be different information about the patient's condition. Could you describe what you noticed about their energy level and overall appearance this morning?"

# 5. State Management

**This section explains the State Management components** (referring to the purple boxes in the main diagram). These components track conversation state and manage pending actions that require user responses.

## 5.1 Pending Actions System

This section explains how the system tracks and resolves pending actions like consent requests. The main diagram shows "State Manager Check" and "Action Response Handler" as purple boxes, but this section details what happens inside those boxes.

**How Pending Actions are Stored:** Pending actions are stored in the database with the following structure:

```
None
{
  "action_id": "uuid",
  "user_id": "user_who_must_respond",
  "circle_id": "care_circle_context",
  "action_type": "share_consent",
  "details": {
    "originatingUser": "who_wants_to_share",
    "target": "who_will_receive_info",
    "reason": "why_sharing_is_needed",
    "content_to_share": "what_will_be_shared",
    "full_question": "complete_question_text_for_matching"
  },
  "created_at": "timestamp",
  "status": "pending"
```

```
}
```

**How User Responses are Parsed:** When a user responds to a pending action, the system:

1. **Intent Classification**: Uses AI to determine if the response is "yes", "no", or "unclear"
2. **Action Matching**: Compares the response against the stored `full_question` text
3. **Context Analysis**: Considers the user's response in context of the original question

**Example Parsing:**

- Question: "Would you like me to share information about your medication adherence with your doctor?"
- User Response: "I can't right now" → Parsed as "no" (decline)
- User Response: "Sure, go ahead" → Parsed as "yes" (approve)
- User Response: "What information exactly?" → Parsed as "unclear" (needs clarification)

**How the Decision is Made:** The decision-making process uses the `matchResponseToAction` function:

1. **Question Matching**: Compares user response against the `full_question` stored in the action
2. **Intent Analysis**: Uses AI to classify the user's intent (approve/decline/clarify)
3. **Confidence Scoring**: Determines how confident the system is in the interpretation
4. **Action Resolution**: If confidence is high, resolves the action; if low, asks for clarification

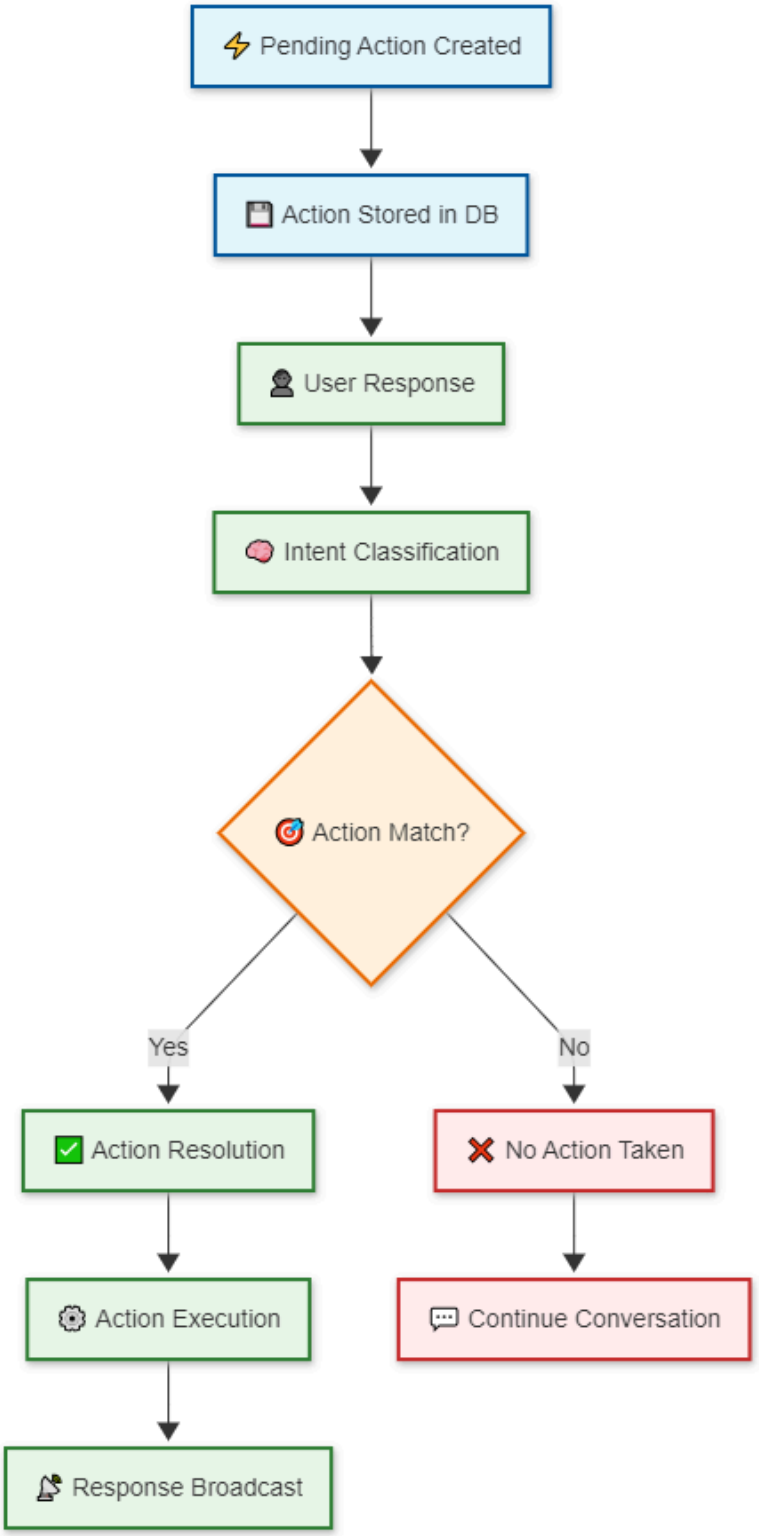



**Fig. 22**: **Pending Action Flowchart**: A flowchart illustrating the state management sub-system for resolving a pending action based on user response and intent classification.

## 5.2 Context Switching

This section explains how users can move between different care circles (e.g., from their own circle to their spouse's circle), which changes which agent they interact with (PersonalAgent vs CareAgent).

**Context Switching Process:**

1. **Circle Validation**: System checks if the user has permission to access the requested care circle

2. **Agent Identity Update**: Changes the user's active agent from PersonalAgent to CareAgent (or vice versa)
3. **Role Assignment**: Updates the user's role context (patient vs caregiver)
4. **Context Confirmation**: Provides feedback to the user about the successful switch

**Grouped Steps**: The context switch involves multiple related operations that happen together:

- Agent Identity Update + Role Assignment + Context Confirmation are all part of a single context switch operation

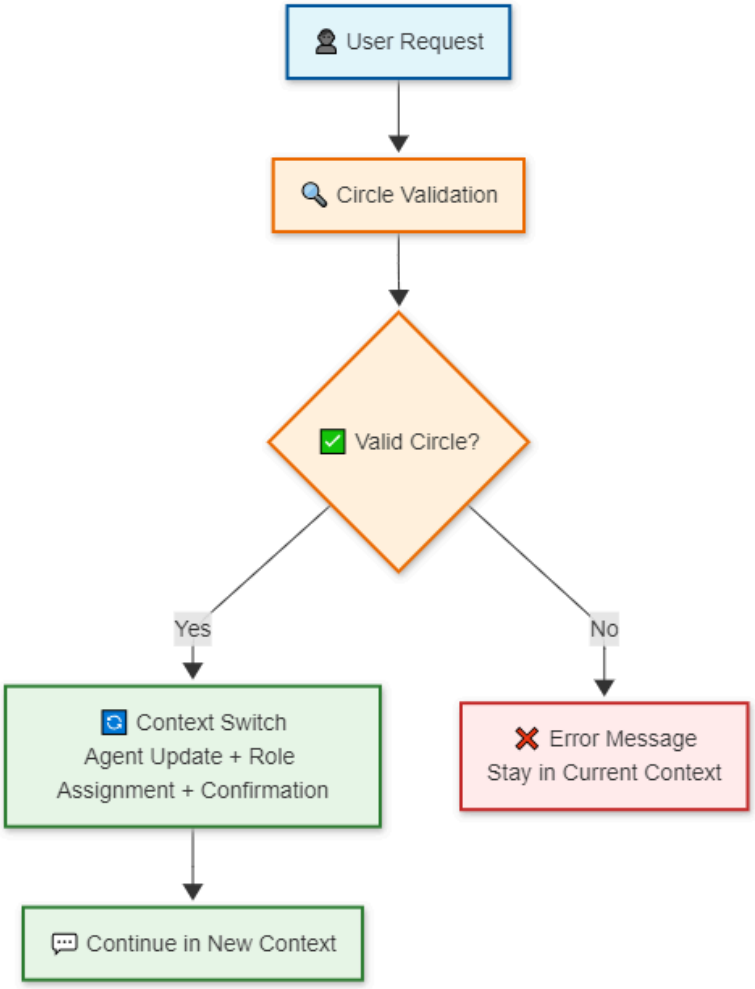
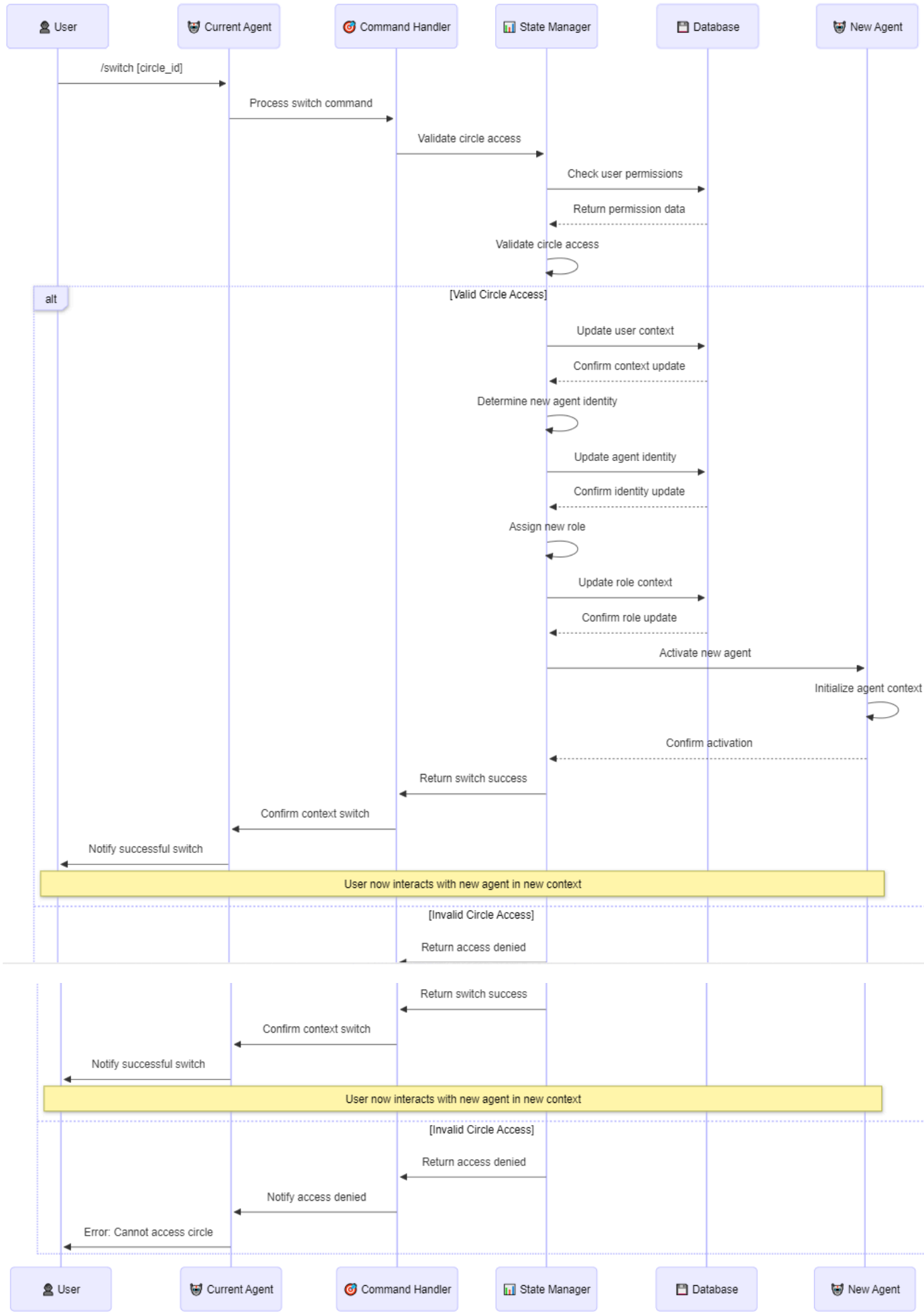



**Fig. 23**: **Context Switching Flowchart**: A flowchart of the high-level logic for the context switching process, from user request to validation and execution.




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**Fig 24**: This diagram shows the context switching process when users move between different care circles, changing their agent identity and role context.

## Key Interactions

### 1. Context Switch Request

- User sends `/switch [circle_id]` command to their Current Agent
- Current Agent forwards the command to Command Handler
- Command Handler delegates to State Manager for validation

### 2. Circle Access Validation

- State Manager validates circle access permissions
- Queries database for user permissions
- Database returns permission data
- State Manager validates if user can access the requested circle

### 3. Valid Access Path

**If the user has valid access to the circle:**

#### Context Update

- State Manager updates user context in database
- Database confirms context update
- User's active circle is changed

#### Agent Identity Update

- State Manager determines new agent identity
- PersonalAgent if user is the primary patient
- CareAgent if user is a caregiver
- Updates agent identity in database
- Database confirms identity update

#### Role Assignment

- State Manager assigns new role based on circle context
- Patient role if user is the primary patient
- Caregiver role if user is a caregiver
- Updates role context in database
- Database confirms role update

#### Agent Activation

- State Manager activates the new agent

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- New Agent initializes agent context
- New Agent confirms activation
- State Manager returns switch success to Command Handler
- Command Handler confirms context switch to Current Agent
- Current Agent notifies User of successful switch

#### 4. Invalid Access Path

##### If the user doesn't have access to the circle:

- State Manager returns access denied to Command Handler
- Command Handler notifies Current Agent of access denial
- Current Agent sends error message to User
- User remains in their current context

### Component Roles

- User: User requesting context switch
- Current Agent: User's current active agent (PersonalAgent or CareAgent)
- Command Handler: Processes switch commands
- State Manager: Manages user context and agent identity
- Database: Stores user permissions, context, and agent identity
- New Agent: User's new agent after context switch

### Context Types

#### Personal Context

- Agent: PersonalAgent
- Role: Patient
- Circle: User's own care circle
- Permissions: Full access to own health information
- Responsibilities: Managing own health and care

#### Care Context

- Agent: CareAgent
- Role: Caregiver
- Circle: Someone else's care circle
- Permissions: Limited access based on consent
- Responsibilities: Supporting patient's care

## Agent Identity Logic

### PersonalAgent

- Trigger: User is the primary patient in the circle

- Context: User's own care circle
- Role: Patient
- Capabilities: Full access to own health information
- Responsibilities: Managing own health and care

### CareAgent

- Trigger: User is a caregiver in the circle
- Context: Someone else's care circle
- Role: Caregiver
- Capabilities: Limited access based on consent
- Responsibilities: Supporting patient's care

### Data Flow

1. Switch Command: User requests context switch
2. Access Validation: System validates circle access permissions
3. Context Update: User's active circle is updated
4. Agent Identity: New agent identity is determined and updated
5. Role Assignment: New role is assigned based on circle context
6. Agent Activation: New agent is activated and initialized
7. Confirmation: User receives confirmation of successful switch

## Example Scenarios

### Patient to Caregiver Switch

- User: Patient in their own circle (PersonalAgent)
- Command: `/switch spouse_circle_id`
- Result: Switches to CareAgent in spouse's circle
- New Role: Caregiver supporting spouse's care

### Caregiver to Patient Switch

- User: Caregiver in spouse's circle (CareAgent)
- Command: `/switch own_circle_id`
- Result: Switches to PersonalAgent in own circle
- New Role: Patient managing own health

### Invalid Access Attempt

- User: Patient in their own circle
- Command: `/switch stranger_circle_id`
- Result: Access denied, remains in current context
- Error: "Cannot access circle: insufficient permissions"

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### Context Persistence

#### Session Management

- Context Storage: User's current context is stored in database
- Session Tracking: Active sessions are tracked per user
- Context Restoration: Context is restored when user reconnects
- Timeout Handling: Context expires after inactivity

#### State Synchronization

- Real-time Updates: Context changes are synchronized in real-time
- Multi-device Support: Context is consistent across devices
- Conflict Resolution: Handles concurrent context changes
- Backup and Recovery: Context state is backed up and recoverable

# 6. Decision-Making Framework

This section explains the the decision logic of the orchestrator (referring to the green "Orchestrator" box in the main diagram) and agents. This shows how the system decides which handler to use based on the message content. The chosen handler may be a part of an orchestrator or an agent.

## 6.1 Orchestrator Decision Tree

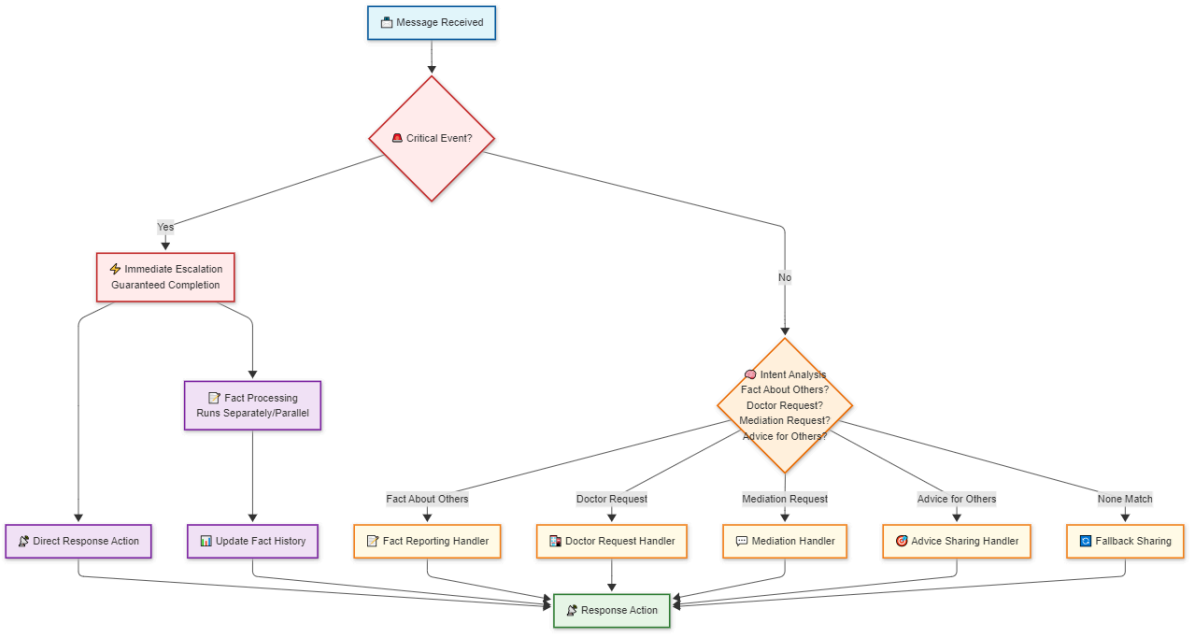



**Fig. 25**: **Orchestrator + Agent decision tree**: A decision tree flowchart illustrating the agent/orchestrator's priority-based routing logic, showing how critical events may bypass certain handlers for reliability

**Decision Tree Logic:** The orchestrator and agents may use a priority-based decision tree:

1. **Critical Events First**: HIGH criticality events take a direct reliable path to ensure direct completion.
2. **Complex Handler Bypass**: Critical events skip logic of complex non-essential handlers to improve reliability and speed

3. **Essential Processing**: Fact processing and other essential processes run separately/parallel for critical events to capture important data
4. **Intent-Based Routing**: Non-critical messages are routed based on detected intents
5. **Fallback Handling**: Messages that don't match specific intents use the standard sharing flow

**Reliability Design:** Different handlers work differently - critical path is simplified for reliability while complex handlers are avoided for critical events.

**Grouped Operations**: Multiple decision points are grouped together as they represent a single analysis phase:

- Fact About Others + Doctor Request + Mediation Request + Advice for Others are all part of the same intent analysis

# 6.2 Consent Management

This section explains the consent logic that runs within multiple handlers, showing how the system determines whether to share information directly or ask for permission first.

**Consent Decision Process:**

1. **Implied Consent Check**: System first checks if user has given implied consent for similar topics
2. **Explicit Consent Level**: If no implied consent, checks user's explicit consent level (none, limited, full)
3. **Consent Request**: If neither implied nor full consent exists, creates a pending action for explicit consent
4. **User Response**: User responds to consent request (approve/decline/clarify)
5. **Action Execution**: Based on response, either shares information or denies the request

**Grouped Operations**: Consent checking and sharing are grouped as they represent a single decision flow:

- Implied Consent Check + Explicit Consent Level Check are part of the same consent evaluation
- Consent Request + User Response + Action Execution are part of the same consent workflow

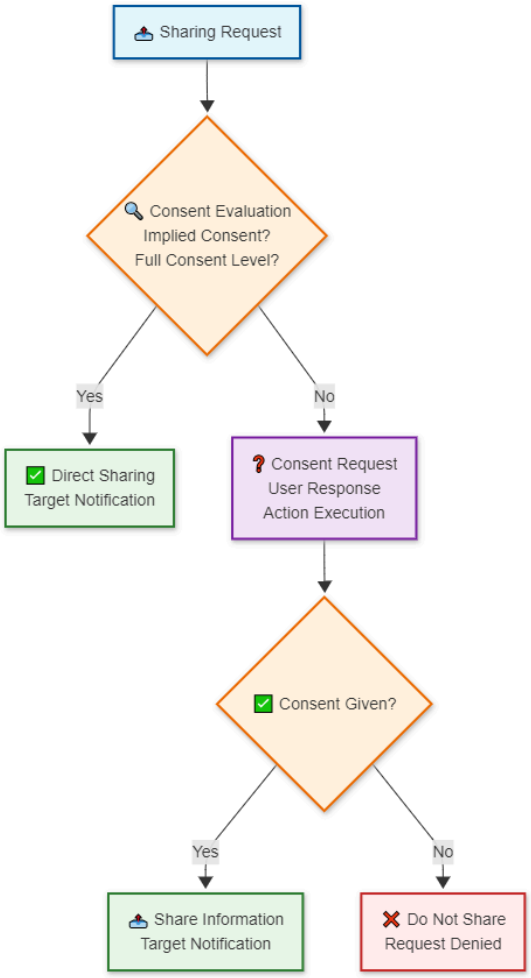



**Fig. 26**: **Consent Management Flowchart**: A flowchart detailing the consent management logic, showing the evaluation of implied and explicit consent before a sharing action is executed.

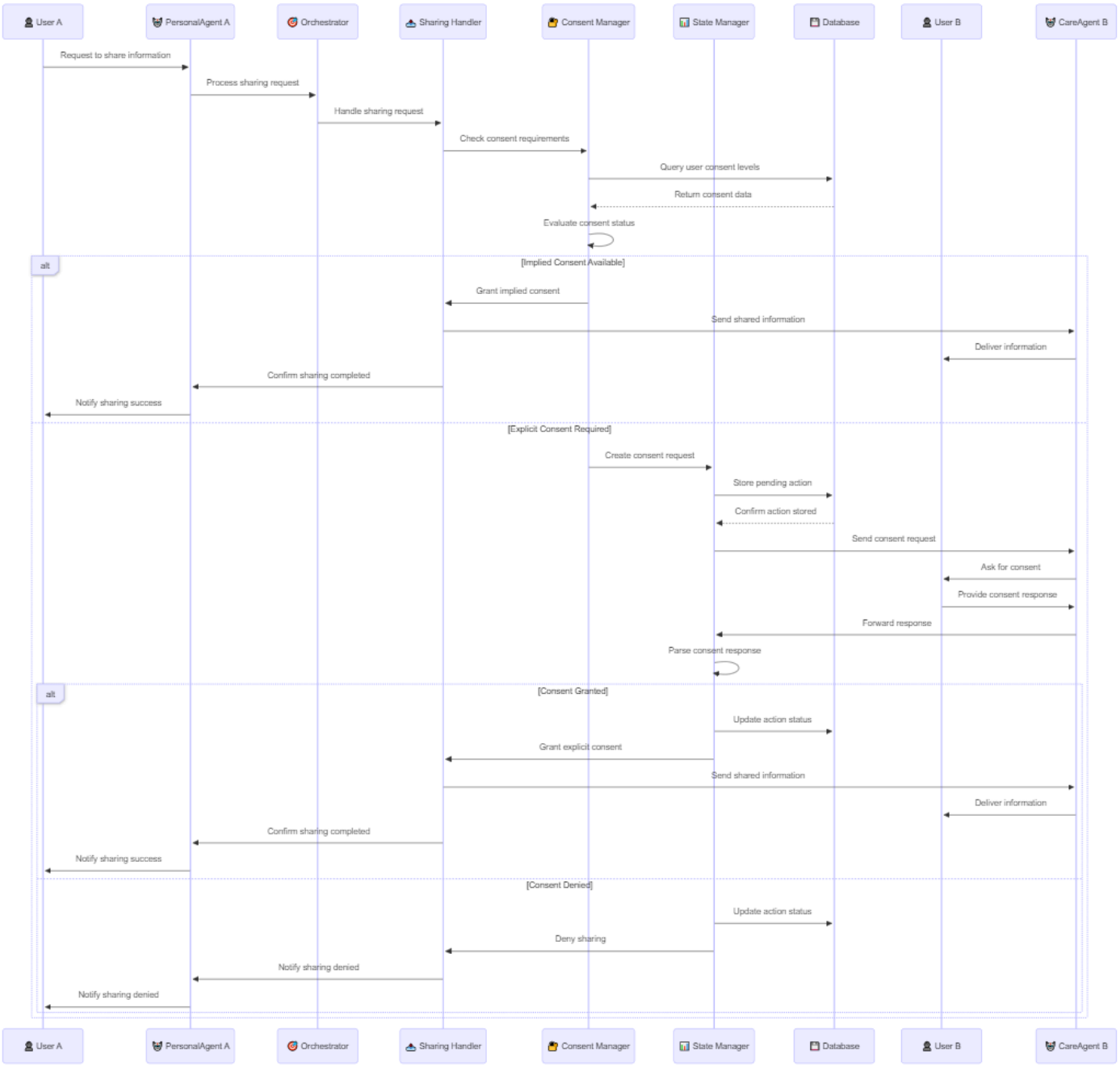



**Fig 27**: This diagram shows the consent management process when users want to share information with others in their care circle.

## Key Interactions

### 1. Sharing Request

- User A requests to share information with someone in their care circle
- PersonalAgent A forwards the request to the Orchestrator
- Orchestrator delegates to Sharing Handler

### 2. Consent Evaluation

- Sharing Handler delegates to Consent Manager
- Consent Manager queries database for user consent levels

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- Database returns consent data (implied, explicit, none)
- Consent Manager evaluates the consent status

**3. Implied Consent Path**

- If implied consent is available, Consent Manager grants consent
- Sharing Handler sends information directly to CareAgent B
- CareAgent B delivers information to User B
- Sharing Handler confirms sharing completion to PersonalAgent A
- PersonalAgent A notifies User A of successful sharing

**4. Explicit Consent Path**

- If explicit consent is required, Consent Manager creates a consent request
- State Manager stores the pending action in database
- State Manager sends consent request to CareAgent B
- CareAgent B asks User B for consent

**5. Consent Response Processing**

- User B provides consent response to their agent
- CareAgent B forwards response to State Manager
- State Manager parses the consent response

**6. Consent Decision**

- If consent granted: State Manager updates action status and grants explicit consent
- Sharing Handler sends information to CareAgent B
- CareAgent B delivers information to User B
- Sharing Handler confirms sharing completion to PersonalAgent A
- PersonalAgent A notifies User A of successful sharing
- If consent denied: State Manager updates action status and denies sharing
- Sharing Handler notifies PersonalAgent A of denial
- PersonalAgent A notifies User A that sharing was denied

## Component Roles

- User A: User requesting to share information
- PersonalAgent A: User A's personal AI agent
- Orchestrator: Central decision-making component
- Sharing Handler: Manages information sharing requests
- Consent Manager: Evaluates consent requirements and status
- State Manager: Manages pending consent actions
- Database: Stores consent levels, pending actions, and action status
- User B: User who must provide consent
- CareAgent B: User B's care agent (when User B is in caregiver role)

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## Consent Levels

### Implied Consent

- Automatic Sharing: Information is shared without explicit permission
- Topic-Based: Consent is implied for specific topics (e.g., medication adherence)
- Relationship-Based: Consent is implied based on relationship (e.g., spouse, primary caregiver)
- Context-Based: Consent is implied based on situation (e.g., emergency situations)

### Explicit Consent

- **Permission Required: User must explicitly approve sharing**
- Topic-Specific: Consent is required for specific topics
- Time-Limited: Consent may have expiration dates
- Revocable: User can withdraw consent at any time

### No Consent

- Sharing Denied: Information cannot be shared
- Privacy Protection: User's privacy is protected
- Opt-Out: User has explicitly denied sharing

## Data Flow

1. Sharing Request: User A's request to share information
2. Consent Check: Evaluation of consent requirements
3. Consent Decision: Implied consent granted or explicit consent required
4. Consent Request: Pending action created (if explicit consent needed)
5. User Response: User B's consent decision
6. Action Execution: Information shared or sharing denied
7. Confirmation: Result communicated to User A

## Example Scenarios

### Implied Consent Example

- User A: "Share my medication adherence with my doctor"
- System: Checks if User A has implied consent for medication topics
- Result: If implied consent exists, information is shared directly

### Explicit Consent Example

- User A: "Share my mental health status with my spouse"
- System: Creates consent request for User B (spouse)
- User B: "Yes, I want to know about their mental health"
- Result: Information is shared with User B

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**Consent Denial Example**

- User A: "Share my medication adherence with my neighbor"
- System: Creates consent request for User B (neighbor)
- User B: "No, I don't want to know about their medication"
- Result: Information is not shared, User A is notified of denial

# 7. Agent Communication Patterns

This section explains how different agents communicate with each other (referring to the overall system flow from the main diagram). It shows the patterns of information flow between users and their respective agents.

## 7.1 Inter-Agent Communication

This section shows the complete communication flow between users and their agents, demonstrating how the system facilitates mediated communication between care circle members.

**Communication Flow:**

1. **User A Message**: Patient or caregiver sends a message to their agent
2. **Agent Processing**: PersonalAgent or CareAgent processes the message through the orchestrator
3. **Analysis & Decision**: System analyzes the message and decides on appropriate actions
4. **Agent-to-Agent Communication**: Information flows from one agent to another (e.g., PersonalAgent to CareAgent)
5. **User B Notification**: Target user receives mediated information through their agent

**Grouped Operations**: Analysis, decision, and action are grouped as they represent a single processing phase:

- Analysis + Decision + Action are all part of the same message processing workflow
- Conflict Detection + Mediated Question + Resolution are part of the same conflict resolution workflow

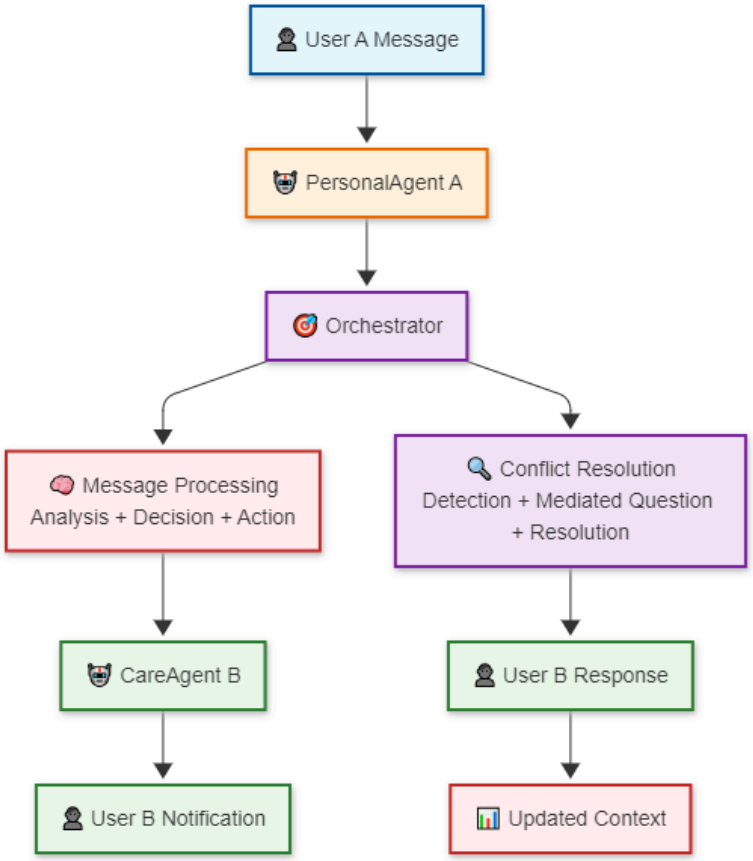



**Fig. 28: Inter-Agent Communication Flowchart**: a flowchart showing the high-level communication patterns between agents for both standard message processing and conflict resolution.

## 7.2 Escalation Flow

This section shows the specific escalation process that occurs when the "Critical Events Handler" detects HIGH criticality events, demonstrating how the system ensures urgent health issues reach the right people quickly.

**Escalation Process:**

1. **Critical Event Detection**: System identifies HIGH criticality health events
2. **Escalation Creation**: Creates escalation record in database
3. **Multi-Channel Notification**: Simultaneously notifies doctor, family members, and updates dashboards
4. **User Confirmation**: Provides feedback to the reporting user
5. **Status Tracking**: Updates system status and logs all actions

**Grouped Operations**: Notification and logging are grouped as they represent parallel processes:

- Doctor Notification + Family Notification + User Confirmation are part of the same notification workflow
- Escalation Log + Doctor Dashboard + Caregiver Alerts + Status Update are part of the same logging workflow

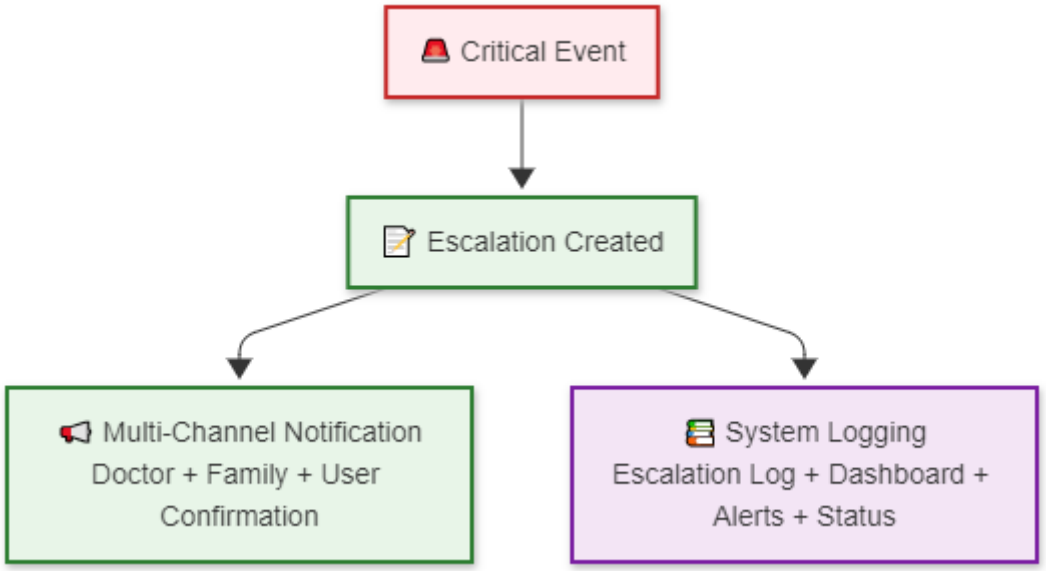



**Fig 29: Escalation Flowchart:** A flowchart illustrating the parallel processes of multi-channel notification and system logging that occur after a critical event is detected.

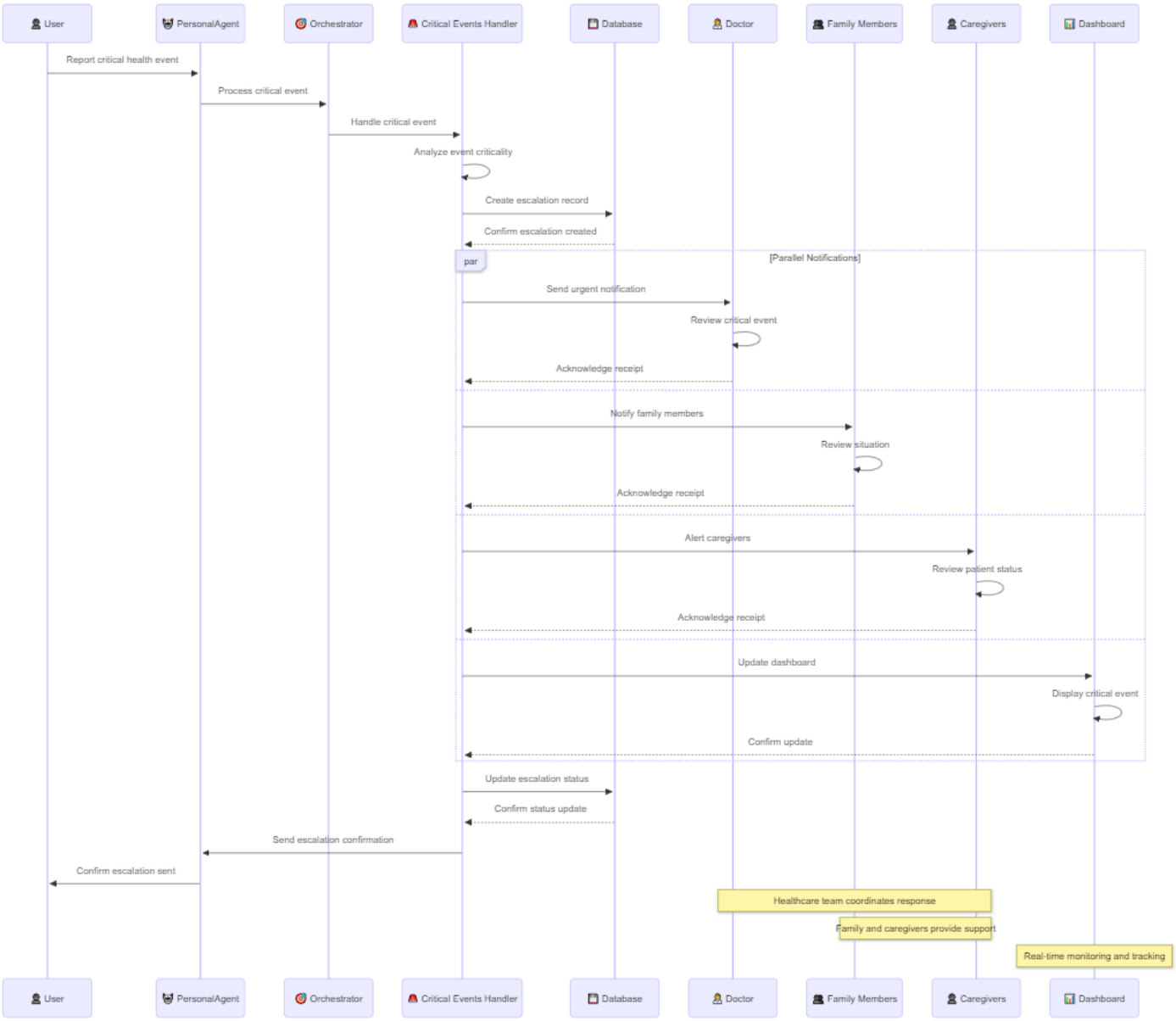



**Fig 30**: This diagram shows the escalation process when critical health events are detected and need immediate attention from healthcare providers and family members.

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## Key Interactions

### 1. Critical Event Detection

- **User** reports a critical health event to their **PersonalAgent**
- **PersonalAgent** forwards the event to the **Orchestrator**
- **Orchestrator** delegates to **Critical Events Handler**

### 2. Event Analysis and Escalation Creation

- **Critical Events Handler** analyzes the event criticality
- Creates escalation record in database
- Database confirms escalation creation

### 3. Parallel Notification System

The system simultaneously notifies multiple parties:

#### Doctor Notification

- **Critical Events Handler** sends urgent notification to **Doctor**
- **Doctor** reviews the critical event
- **Doctor** acknowledges receipt of notification

#### Family Notification

- **Critical Events Handler** notifies **Family Members**
- **Family Members** review the situation
- **Family Members** acknowledge receipt of notification

#### Caregiver Alert

- **Critical Events Handler** alerts **Caregivers**
- **Caregivers** review patient status
- **Caregivers** acknowledge receipt of alert

#### Dashboard Update

- **Critical Events Handler** updates the **Dashboard**
- **Dashboard** displays the critical event
- **Dashboard** confirms the update

### 4. Status Tracking and Confirmation

- **Critical Events Handler** updates escalation status in database
- Database confirms status update
- **Critical Events Handler** sends escalation confirmation to **PersonalAgent**
- **PersonalAgent** confirms escalation sent to **User**

### 5. Coordinated Response

- **Healthcare team** coordinates response based on criticality
- **Family and caregivers** provide support and monitoring
- **Dashboard** provides real-time monitoring and tracking

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## Component Roles

- **User**: Patient or caregiver reporting the critical event
- **PersonalAgent**: User's personal AI agent
- **Orchestrator**: Central decision-making component
- **Critical Events Handler**: Manages critical event detection and escalation
- **Database**: Stores escalation records and status updates
- **Doctor**: Healthcare provider receiving urgent notifications
- **Family Members**: Family members who need to be informed
- **Caregivers**: Caregivers who need to be alerted
- **Dashboard**: Real-time monitoring and tracking system

## Criticality Levels

### HIGH Criticality

- **Immediate Escalation**: Bypasses all other logic
- **Multi-Channel Notification**: Doctor, family, caregivers, dashboard
- **Urgent Response**: Requires immediate attention
- **Status Tracking**: Continuous monitoring until resolved

### MEDIUM Criticality

- **Standard Escalation**: Follows normal escalation process
- **Selective Notification**: Based on event type and context
- **Timely Response**: Requires attention within reasonable time
- **Periodic Updates**: Regular status updates

### LOW Criticality

- **Information Sharing**: Standard information sharing process
- **Optional Notification**: May or may not require notification
- **Routine Response**: Standard care process
- **Basic Tracking**: Minimal status tracking

## Data Flow

1. **Critical Event**: User reports critical health event
2. **Event Analysis**: System analyzes event criticality
3. **Escalation Creation**: Escalation record created in database
4. **Parallel Notifications**: Multiple parties notified simultaneously
5. **Status Updates**: Escalation status tracked and updated
6. **Confirmation**: User receives confirmation of escalation
7. **Coordinated Response**: Healthcare team coordinates response

## Example Scenarios

### Chest Pain Escalation

- **User**: "My chest hurts. I want to talk to my doctor"
- **System**: Detects HIGH criticality for chest pain
- **Escalation**: Doctor notified immediately, family alerted, dashboard updated
- **Response**: Doctor contacts patient, family provides support

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#### Fall Incident Escalation

- **User**: "A fell down from the 3rd floor!"
- **System**: Detects HIGH criticality for fall incident
- **Escalation**: Emergency services notified, family alerted, caregivers informed
- **Response**: Emergency response coordinated, family provides support

#### Medication Error Escalation

- **User**: "I think I took the wrong medication"
- **System**: Detects MEDIUM criticality for medication error
- **Escalation**: Doctor notified, family informed, dashboard updated
- **Response**: Doctor provides guidance, family monitors situation

### Escalation Tracking

#### Real-Time Monitoring

- **Dashboard Updates**: Continuous status updates
- **Response Tracking**: Monitor response times and actions
- **Status Changes**: Track escalation status changes
- **Resolution Tracking**: Monitor until issue is resolved

#### Communication Logs

- **Notification History**: Track all notifications sent
- **Response Logs**: Record responses from healthcare team
- **Status Updates**: Log all status changes
- **Resolution Notes**: Document resolution proces

# 8. Diagram Navigation Guide

## 8.1 How to Use This Document

**Main Diagram Reference**: The main "Message Processing Flow" diagram (Section 3.1) serves as the central reference point. Each subsequent section explains what happens inside specific boxes from this main diagram:

- **Section 3.2**: Explains the red "AI Analysis Handler" box
- **Section 4**: Explains the yellow "Sharing Handler" and "Conflict Handler" boxes
- **Section 5**: Explains the purple "State Manager Check" and "Action Response Handler" boxes
- **Section 6**: Explains the green "Orchestrator" decision logic
- **Section 7**: Shows overall communication patterns between all components

## 8.2 Color Coding Legend

Throughout all diagrams, colors represent different types of components:

- 🔵 **Blue**: User input and user interactions
- 🟢 **Green**: Orchestration, output, and successful operations
- 🟠 **Orange**: Decision points and command processing
- 🟣 **Purple**: State management and data storage
- 🔴 **Red**: AI processing, critical events, and error states
- 🟡 **Yellow**: Business logic handlers and specialized processors

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## 8.3 Component Type Indicators

Each diagram uses emojis to quickly identify component types:

- 👤 **Users**: Human participants in the system
- 🤖 **Agents**: AI conversational entities
- ⚙️ **Handlers**: Business logic processors
- 🧠 **AI Processing**: LLM analysis and decision making
- 📊 **State Management**: Conversation state tracking
- 🎯 **Orchestration**: Central workflow coordination
- 📡 **Output**: Response delivery and notifications

This framework provides a robust foundation for the multi-agent healthcare system, ensuring clear responsibilities, effective communication, and reliable decision-making processes.

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## Annexure- I

## *Illustrated examples for mentioning use case of Product/ Process*

1. ***"Classification of Hard and Soft Taps on Capacitive Touch Screen" having application in below mentioned use case:***

   *We all use capacitive touch screens, on a series of devices starting with smart watches, smartphones tablets laptops and desktops. The only thing you can do on the present touch screens is to indicate a location, by touching it there. It doesn't matter whether you touch it hard or lightly - the touch is definitely not 3D. iPhone X has atte mpted to introduce this feature (force touch) by including expensive special pressure-sensitive hardware. We adopt a different approach, that requires you to only install an app on the existing device. With this the same touch screen that you have been using now becomes sensitive to at least two levels of pressure, light and heavy. Every other app developer can now exploit this feature and provide more sophisticated user interfaces which can distinguish levels of touch. The possibilities are limited only by the imagination. As the technology matures, a larger number of levels of touch is likely to be supported.*

2. ***"Antibacterial Nano breathing Nasal Filter" having application in***

   *Many people use face mask for breathing pollution free air, but the main constraint in breathing is decrease in breathing flow rate. The innovation described herein relates to a nasal air filter, and more particularly, the invention relates to a nasal air filter based on nanotechnology for breathing by human beings that has antibacterial feature, which can mimic the natural breathing process i.e. 12-15 lpm and is comfortable to use. The nasal filter will be useful for people living in polluted cities as well persons having allergy and suffering from Asthama & Bronchitis.*

3. ***"A method of measuring BMP signaling using BMP responsive reported cell line" having application in***

   Bone Morphogenetic Protein (BMP) signaling is necessary and sufficient for bone formation.

   It is present in several biological samples measurement of which may have diagnostic value. However, at present there is no sensitive method of detecting BMP proteins in a biological sample. In this disclosure we describe creation of a cell line based sensitive and accurate method of estimating BMP proteins in any specimen.

4. ***A unique device for plasma processing to simulating magnetospheres in the laboratory***

   *The dipole plasma device would be helpful in industry for plasma processing of samples such as in the semiconductor industry, where energetic electrons (or ions) are required to impinge on a substrate and to bring about desired changes in the substrate such as plasma assisted ion doping, etching, or creation of nanostructures on metallic surfaces, the device would be helpful in understanding the physics of plasmas confined in a magnetic dipole.*

5. ***Large area micro-texturing on free-form surfaces using flexible-electrode throughmask electrochemical machining***

   Surface micro-texturing deals the issues pertaining to various fields of engineering for enhancing the essential functions such as tribological, wetting, biocompatibility, sustainability, cleanliness etc. Among all the aforementioned sectors, micro-texturing of free-form large areas is getting huge attention, e.g. micro-textures on artificial biomedical implants enhances sustainability and life cycle by better implant-tissue interface,

cell-adhesion and cell proliferation. Micro-textures on cylindrical surfaces (both internal and external) of bearings, piston rings, hypodermic needles assists in reducing the coefficient of friction and facilitating lubrication.

**This page is for reference purpose, no need to print this page.*